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Article

New U–Pb geochronologic and palaeomagnetic constraints on the late Palaeoproterozoic Hartley magmatic event: evidence for a potential large igneous province in the Kaapvaal Craton during Kalahari assembly, South Africa

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Abstract: The volcanic Hartley Formation (part of the Olifantshoek Supergroup, which is dominated by red bed successions) in South Africa recorded depositional and tectonic conditions along the western Kaapvaal Craton during the late Palaeoproterozoic. It formed in association with red bed deposition elsewhere in the cratonic hinterland and along the craton's northern margin. However, the exact correlation of the Olifantshoek Supergroup with these other red-bed successions is hindered by poor geochronological constraints. Herein, we refine the age and palaeopole of the Hartley Formation, and provide geochronological constraints for large-scale 1.93–1.91 Ga bimodal magmatism on the Kaapvaal Craton (herein named the Hartley large igneous province). We present new age constraints for the mafic and felsic phases of this event at 1923 ± 6 Ma and 1920 ± 4 Ma, respectively, which includes the first reported age dating of the Tsineng Dyke Swarm that has been linked to Hartley volcanism. A mean 1.93–1.91 Ga palaeomagnetic pole for the Hartley large igneous province at 22.7°N, 328.6°E with A_{95} = 11.7° represents a significant improvement on a previously published virtual geomagnetic pole. This improved pole is used to refine the late Palaeoproterozoic apparent polar wander path of the Kaapvaal Craton. This can assist in correlation of red-bed successions in southern Africa.

Keywords: geochronology; large igneous province; dyke swarm; palaeomagnetism; apparent polar wander path; Orosirian period

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1. Introduction

The identification and understanding of Mesoarchean to Proterozoic magmatic events on the Kaapvaal Craton, including short-lived large igneous provinces (LIPs), are becoming better understood using high-precision U–Pb geochronology on baddeleyite and zircon (e.g., Olsson et al. 2010, 2011; Gumsley et al. 2013, 2015a; Mukasa et al. 2013; de Kock et al. 2014; Kampmann et al. 2015). The late Palaeoproterozoic Hartley Formation represents a small volcanic component within the red bed-dominated Olifantshoek Supergroup (Cornell 1987; Moen 1999, 2006), which in southern Africa recorded conditions along the western Kaapvaal Craton margin (Fig. 1). Several magmatic rock units on the Kaapvaal Craton have previously been correlated to the Hartley Formation (Cornell et al. 1998; Hanson et al. 2004; Goldberg 2010), and hint at a wider spatial distribution of magmatism, but the association is as yet unconfirmed for most. The Olifantshoek Supergroup developed in broad association with red bed deposition elsewhere in the cratonic interior following the emplacement of the ca. 2.05 Ga Bushveld Complex and 2.04–2.02 Ga high-grade metamorphism during the assembly of the Kalahari Craton (Fig. 1). Thin-skin tectonic shortening of much of the Olifantshoek Supergroup occurred during the <1.93 Ga Kheis Orogeny (Cornell et al. 1998), and later tectonic reworking by the ca. 1.0 Ga Namaqua-Natal Orogeny (Eglington & Armstrong 2004). Structural complexity, younger cover and limited geochronology obstruct the exact correlation of the Olifantshoek Supergroup with other southern African red-bed suc-

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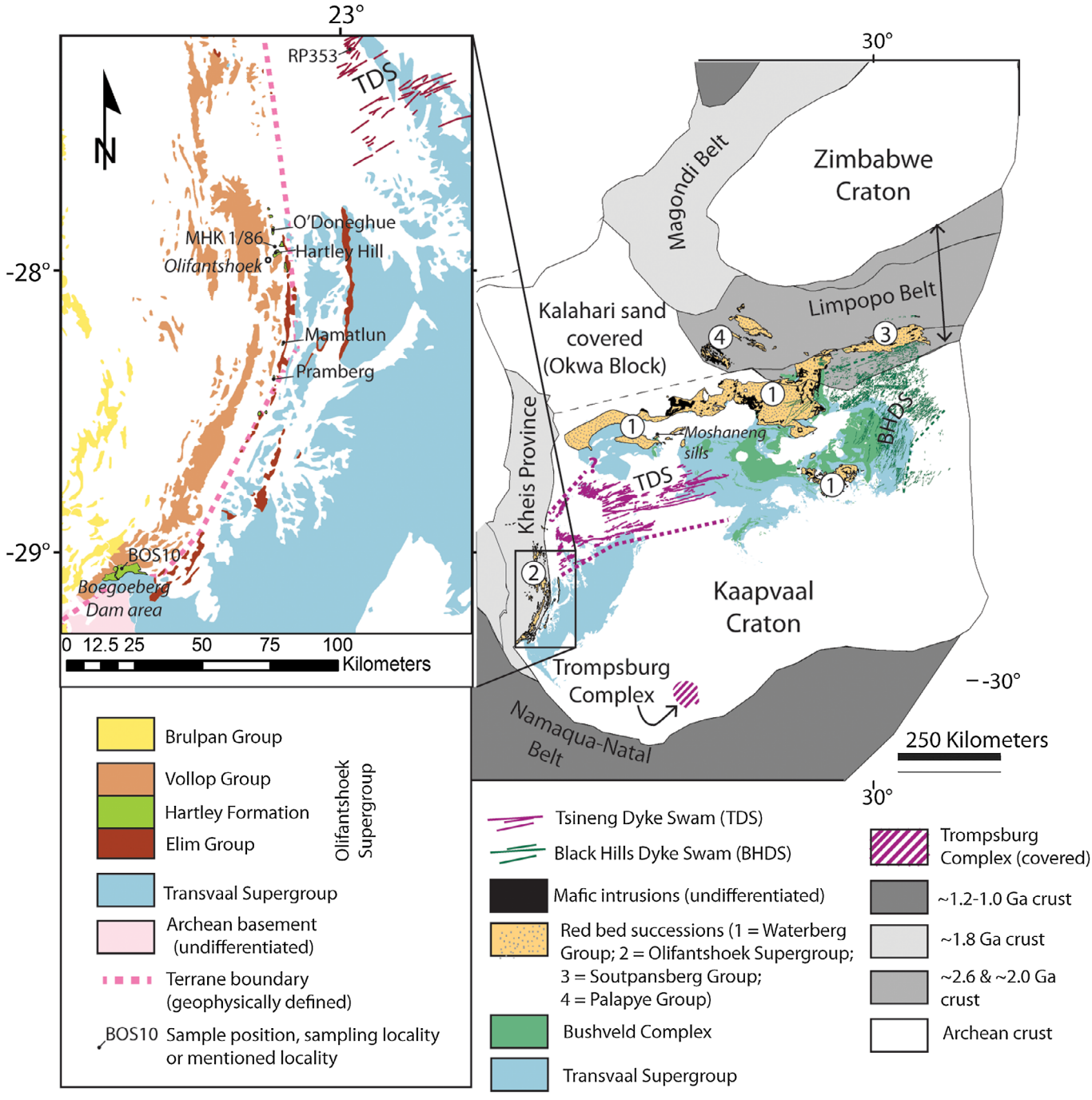


Fig. 1. An overview map of the Kaapvaal Craton and its surrounding environment which highlights the Olifantshoek-Waterberg-Soutpansberg-Palapye strata and the ~1.93–1.84 Ga dolerite dyke and sill intrusions. Inset shows a geological map of the western margin of the Kaapvaal Craton. Sample localities for palaeomagnetism and geochronology near the western margin of the craton are shown together with other localities mentioned in the text.

cessions. Previous attempts to date the Hartley Formation document imprecise Rb–Sr and Pb–Pb isochrons and errorchrons (Armstrong 1987; Cornell 1987). An exception is a $^{207}Pb/^{206}Pb$ age of 1928 ± 4 Ma for a quartz porphyry unit interbedded with the Hartley Formation basalts obtained by Cornell et al. (1998) via Pb–Pb evaporation. This provides the principal constraint on the Kheis Orogeny. Here we present a U–Pb SHRIMP zircon age of another porphyry occurrence that confirms the age for the Hartley Formation. U–Pb age constraints from two suggested Hartley Formation equivalents are reported to constrain the regional extent and establish LIP status of Hartley volcanism during the Kalahari Craton assembly. Coupled with a refinement of previous palaeomagnetic studies (Evans et al. 2002), our new ages further constrain the apparent polar wander (APW) path of the Kaapvaal and Kalahari Cratons during the Orisirian Period – a vital tool in the correlation of southern African red-bed successions.

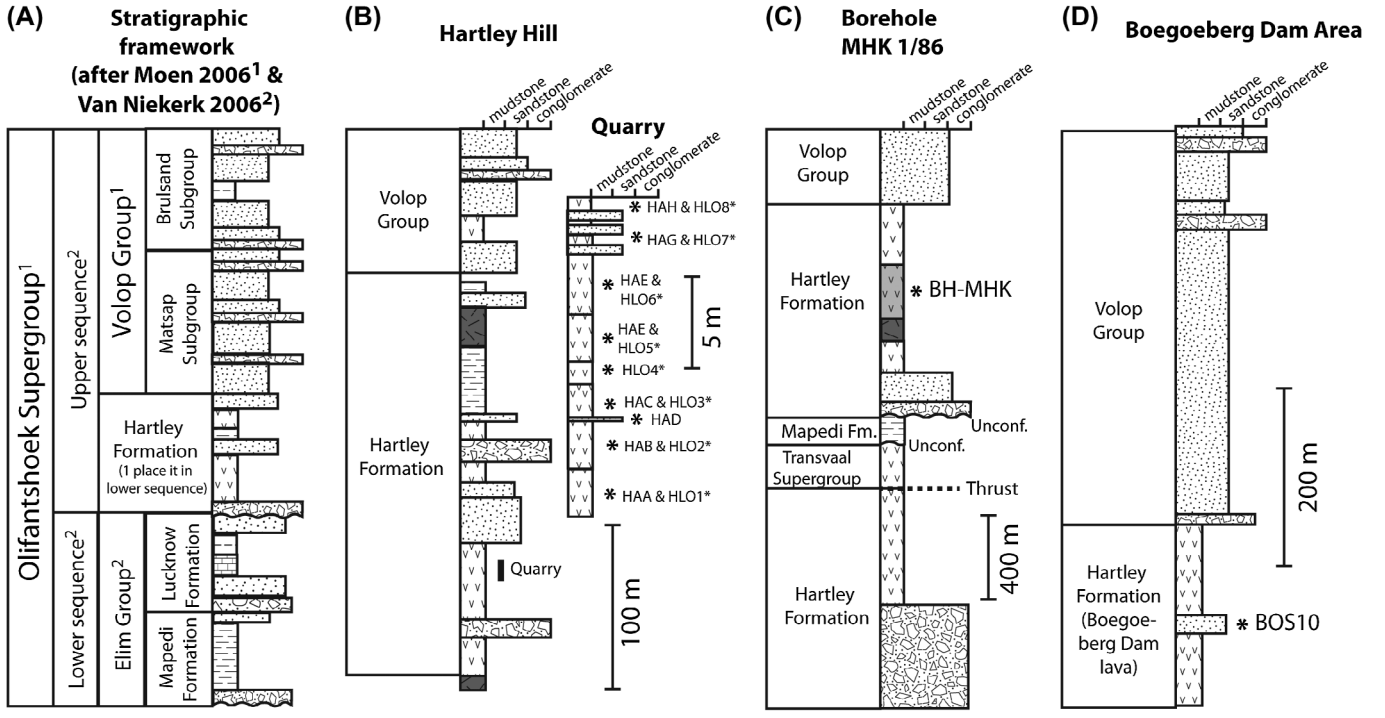


Fig. 2. Lithostratigraphy of the Olifantshoek Supergroup. **A**. Stratigraphic framework of the Olifantshoek Supergroup after Moen (2006) and Van Niekerk (2006) used in this study. **B.** Lithostratigraphy at Hartley Hill (modified after Cornell 1987). **C.** Lithostratigraphy of drill core MHK 1/86 from the farm Mooihoek, north of Olifantshoek, and **D**. Lithostratigraphy at Boegoeberg Dam. Position of U–Pb geochronology samples as well as palaeomagnetic sampling sites are indicated as stars. Asterix after sampling sites indicate sites of Evans et al. (2002).

2. Geological setting

2.1. *Regional geology*

The Kaapval Craton together with the Zimbabwe Craton, and several accreted terranes along their western margins (Eglington & Armstrong 2004), constitute the greater Kalahari Craton. To its north, the Kaapvaal Craton adjoins with the Zimbabwe Craton along the Limpopo Belt (Fig. 1), a 2.6–2.02 Ga polymetamorphic complex (Kramers & Mouri 2011). However, the exact timing of amalgamation of the Kalahari Craton remains controversial (Hanson et al. 2011). On its west, the Kaapvaal Craton is bounded by the Kheis Province (Fig. 1), a <1.93 Ga fold and thrust belt, which has largely been reworked during the ca. 1100 and 1050 Ma Namaqua-Natal Orogeny (Fig. 1), to the west and south of the craton (Eglington & Armstrong 2004; Cornell et al. 2006; Eglington 2006).

During the late Palaeoproterozoic extensive red-bed successions were deposited on the Kaapvaal Craton and its margins including the Limpopo Belt and Kheis Province. The Waterberg Group, Soutpansberg Group and Olifantshoek Supergroup of South Africa together with the Palapye Group (Botswana), and a number of smaller outliers, such as the Blouberg Formation, are erosional remnants of these late Palaeoproterozoic successions (e.g., De Kock et al. 2006; Dorland et al. 2006). It is now accepted that deposition of the Soutpansberg Group commenced some 200 million years after the initiation of deposition in the Waterberg Group and Blouberg Formation at 2.04 Ga (Dorland et al. 2006; Corcoran et al. 2013; Geng et al. 2014), while deposition continued possibly as late as 1604 Ma within the Palapye Group (Mapeo et al. 2004). Still elusive is the exact correlation of these successions with the Olifanshoek Supergroup.

2.2. *Olifanshoek Supergroup*

Structural complexities and poor geochronological control have resulted in several proposed lithostratigraphic frameworks for the Olifantshoek Supergroup. Here, we employ a combination of the recent frameworks of Moen (2006) and Van Niekerk (2006) that sees the Olifantshoek Supergroup divided into two unconformity-bounded sequences (Fig. 2). The oldest of these sequences is named the Elim Group in accordance with Van Niekerk (2006). The formations of this group are interpreted by Van Niekerk (2006) and others (e.g., Beukes et al. 2002; Evans et al. 2002) as being part of the Transvaal Supergroup and not an integral part of the Olifanshoek Supergroup as proposed by Moen (2006). The placement by Moen (2006) seems justified by recent unpublished constraints from detrital zircon grains from the Mapedi Formation (Da Silva 2011). Placement of the Elim Group rocks either before (i.e., Transvaal Supergroup) or after (i.e., Olifantshoek Supergroup) the intrusion of the 2.05 Ga Bushveld Complex is crucial for how the Kaapvaal Craton's APW path is defined, as will be shown (Section 5.3). The Elim Group consists of the Mapedi and Lucknow formations (Moen 1999), an upward coarsening shale to quartzite succession with interbedded carbonate rocks.

As recognized by Van Niekerk (2006), the second sequence is taken from the regional unconformity at the base of the overlying Hartley Formation, which is composed of basal conglomerate and quartzite (i.e., the so-called Neylan Member of the Hartley Formation in the subdivision of Van Niekerk, 2006) followed by dominantly volcanic rocks (Fig. 2). The Hartley Formation is overlain conformably by light grey to white arenites (orthoquartzite), with minor shale and poorly exposed silty sandstone (Fig. 2). Opinions vary widely on the lithostratigraphic status

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of this sandstone-dominated unit at the top of the Volop Group (Moen 2006; Van Niekerk 2006).

2.3. Hartley volcanic rocks and equivalents

The Hartley Formation is formed by dominantly basaltic lava flows also encompassing pyroclastic and sedimentary rocks, with a total estimated thickness that ranges between 300 and 762 m (Cornell 1987). The conservative estimate is supported by reports of thrust duplication (Tinker et al. 2002). Rare quartz-feldspar porphyry also occurs as isolated outcrops in the predominantly Kalahari sand covered N–S trending valleys of the present topography south of Olifantshoek (Cornell 1987). The 1928 ± 4 Ma age of Cornell et al. (1998) is from such porphyry that outcrops on the farm Pramberg (Fig. 1). In this work, another porhyry intersected in a borehole north of Olifantshoek (MHK 1/86), was sampled for U–Pb geochronology.

Cornell et al. (1998) proposed that basalt and interbedded sedimentary units some 140 km south of Olifantshoek (Fig. 1) are Hartley Formation correlatives. This unit (the Boegoeberg Dam Formation) could, however, also represent a volcanic unit of the 2.64 Ga lower Transvaal Supergroup (Smit et al. 1991). Fine-grained quartzite (sample BOS 10), interbedded with the Boegoeberg Dam Formation was sampled to test the proposed correlation.

In addition, an undated ~120 m-thick unit of mafic lava preserved in the lower part of the Selika Formation near the base of the Palapye Group in Botswana (Ermanovics et al. 1978), must be older than 1604 Ma according to Mapeo et al. (2004), and could represent a Hartley Formation equivalent. Elsewhere, near Moshaneng, Hanson et al. (2004) reported a narrow range of ~1927 Ma $^{207}Pb/^{206}Pb$ baddelyite ages for dolerite sills that intrude the Waterberg Group (Fig. 1). They further illustrated geochemical similarity between these sills and basalt of the Hartley Formation.

In South Africa, U–Pb geochronology of zircons from two gabbroic drill core samples of the unexposed Trompsburg Complex near the southwestern Kaapvaal Craton margin, beneath the Phanerozoic cover (Fig. 1), yielded an age of 1915 ± 6 Ma (Maier et al. 2003), which is comparable to the existing ages for the Hartley Formation (Cornell et al. 1998) and Moshaneng sills (Hanson et al. 2004). Finally, a distinct dyke swarm known as the Tsineng Dyke Swarm (TDS), in the western Kaapvaal Craton (Fig. 1), has been linked to the Hartley Formation volcanic rocks (Goldberg 2010). The TDS is evident on aeromagnetic maps, and displays a slight convergence towards the west, where it is truncated by the Kheis Province. This truncation led Goldberg (2010) to suggest a pre-1.8 Ga age for the TDS. The dyke swarm fans out very slightly towards the ENE, and its strike is ~0–70°, with a slight dextral curvature to ~0–60° in the west. According to Goldberg (2010), aeromagnetic data indicate that these dolerite dykes crosscut the ca. 2.4 Ga Ongeluk Formation of the Transvaal Supergroup (Fairey et al. 2013; Gumsley et al. 2015b), presenting a maximum constraint on TDS emplacement. One of the NE-trending TDS dykes intersected in drill core near the town of Tsineng (sample RP353) was collected for U–Pb geochronology.

3. Methodology

3.1. SHRIMP U–Pb geochronology of the Hartley volcanic rocks

Zircon grains were separated from a quartz porphyry found at 683 m depth within drill core of borehole MHK1/86 (Fig. 2), from the farm Mooihoek just north of Olifantshoek (Fig. 1), as well as from fine-grained quartzite (Sample BOS 10) interbedded with volcanic rocks of the Boegoeberg Dam Formation (Figs. 1 and 2). A heavy mineral fraction was extracted using Bromoform on a washed 250 micron sieve after crushing and milling at the University of Johannesburg, South Africa. Zircons were concentrated from this heavy mineral fraction by magnetic separation at 1.25 A, and then separated using methylene iodine in the heavy mineral separation facilities of the PRISE Research School of Earth Sciences at the Australian National University in Canberra, Australia.

For microbeam analysis, the zircon grains were hand-picked, mounted in epoxy resin, polished and gold-coated. Cathodoluminescence and backscatter electron images were obtained by scanning electron microscopy, which were used to identify xenocrysts through morphology, and to avoid inclusions, cracks and metamict zones in grains.

Sensitive high-resolution ion microprobe reverse geometry analyses were performed at PRISE at the Australian National Uuniversity. The FC-1 and SL3 zircon standards were used for calibration (Compston & Williams 1984; Paces & Miller 1993). Common Pb corrections were done using the Stacey & Kramers (1975) terrestrial Pb model composition. The analytical data were reduced according to the method described in Williams & Claesson (1987) and Compston et al. (1992). Only $^{207}Pb/^{206}Pb$ ages with a discordance of less than 10% were used to define detrital zircon age populations in sample BOS10. Uncertainties of ages are given at 2σ in the text, ignoring decay constant errors.

3.2. ID-TIMS U–Pb geochronology of the Tsineng Dyke Swarm

Baddeleyite grains were extracted from drillcore sample RP353, which intersects one of the NE–SW trending TDS dolerite dykes. The borehole is located on Smartt Mine in the Kalahari Manganese Field near Tsineng in the Northern Cape Province of South Africa (see Fig. 1). Here, a dyke intrudes Mn-rich rocks of the Transvaal Supergroup (Hotazel Formation). The RP353 dyke sample is an unfoliated coarse-grained dolerite with abundant plagioclase phenocrysts. Baddeleyite grains were extracted after crushing, milling and water-based separation at Lund University following the procedures described in Schütte (1992). The best-quality grains were hand-picked and transferred to Teflon capsules, where they were washed repeatedly in ultrapure water and dilute HNO_3. One drop of a $^{233-236}U–^{205}Pb$ tracer solution and 10 drops of $HF–HNO_3$ (10:1) solution were added into each capsule. The samples were then completely dissolved after three days in the oven. Isotope Dilution Thermal Ionization Mass Spectrometry (ID-TIMS) analysis was done using a Finnigan Triton Thermal Ionization Mass Spectrometer in the Department of Geosciences at the Swedish Museum of National History in Stockholm. Data were plotted using Isoplot after Ludwig (1991). U decay constants used were those published in Jaffey

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et al. (1971). The correction for initial common Pb was carried out using the isotopic model compositions of global common Pb (Stacey & Kramers 1975). Further details are given in Olsson et al. (2010).

3.3. *Palaeomagnetism of the Hartley volcanic rocks*

Ten individual magmatic cooling units of the Hartley Formation were drilled and sampled for palaeomagnetic study (Fig. 1). A total of 77 oriented samples, with between 8 and 9 samples per unit, were taken using a water-cooled portable drill. The majority of samples originate from the Hartley Hill quarry outside of the town of Olifantshoek, where a largely undeformed succession of Hartley Formation volcanic rocks are exposed as eight sub-horizontal flow units (Cornell 1987). This locality was also previously sampled for palaeomagnetism by Evans et al. (2002). Additional samples were taken from a more foliated mafic volcanic unit on the farm O'Doneghue, and from the dated porphyry of Cornell et al. (1998) outcropping on the farm Pramberg. Sample orientation was done using a magnetic and sun compass, and drilled samples were cut into one or more shorter cylindrical specimens.

In the palaeomagnetic laboratory at UJ, the cut specimens were measured on a vertical 2G Enterprises DC-4 K magnetometer equipped with an automatic sample changer (Kirschvink et al. 2008). After measuring the natural remanent magnetization (NRM) of all specimens, they were demagnetized stepwise. Specimens were pretreated with low-field strength alternating field (AF) demagnetization from 2 mT up to 10 mT. Specimens that showed a strong resistance to this initial low-field strength AF demagnetization were submerged in liquid nitrogen before demagnetization was continued. Specimens were then treated to high-field strength AF demagnetization in steps up to 100 mT. Magnetic components were identified and quantified via the least-squares (LSQ) principal component method of Kirschvink (1980) using Palaeomag 3.1b2 software of Jones (2002). Palaeopoles were plotted using GPlates 1.2.0 software of Williams et al. (2012).

4. Results

4.1. *Zircon U–Pb geochronology*

Summarized geochronological data are presented in Table 1 and Table 2, with concordia diagrams and detrital age populations illustrated in Fig. 3.

4.1.1. *Mooihoek Porphyry (MHK1/86)*

This quartz porphyry yielded clear euhedral zircons with consistent Th/U ratios of between 0.5 and 0.75 indicating their magmatic origin. Seventeen zircon crystals analysed by SHRIMP-RG in 18 spots produce a well-defined U–Pb concordia age of 1920 ± 4 Ma with MSWD of 0.04 (Fig. 3A).

4.1.2. *Boegoeberg Dam Formation (BOS10)*

Sample BOS10 yielded zircon grains of between 75 and 150 μm in length that varied from being euhedral to well rounded. Some grains were clearly broken during the milling process and could have been in excess of 250 μm originally. Most zircon grains were clear, but about 15% of grains were covered in a mottled red coating. About half of the grains were pitted and cracked, while a few grains had inclusions or growth of xenotime along grain boundaries. The zircon grains were concentrically zoned, and some had detrital cores with younger zircon rims.

SHRIMP analyses of 52 zircon grains produced 29 concordant to near concordant (within 10%) ages that define two major age populations (Fig. 3B). All concordant and near-concordant grains from this sample had high Th/U ratios and are considered magmatic in origin. Eleven grains define a prominent ca. 2000–2100 Ma age peak, and 6 grains define a prominent age population around ca. 2700 Ma. There is also a secondary peak consisting of 3 grains around ca. 3200 Ma. Most interesting for our study was one zircon grain with an age of 1931 ± 20 Ma (analysis 39.2). Another zircon grain produced an age of 1957 ± 18 Ma (analysis 33.2) and may fall in the same young group. The two grains are both euhedral and may be derived from the Boegoeberg Dam Formation volcanic eruptions (Table 1).

4.2. *Baddeleyite U–Pb geochronology*

Summarized geochronological data are presented in Table 2, with a concordia diagram shown in Fig. 3. Baddeleyite grains extracted from the dolerite dyke drillcore sample (RP353) are transparent, light brown crystals with no signs of mottled surfaces or alteration. Three grain fractions plot concordant, yielding a robust concordia age of 1923 ± 6 Ma (MSWD = 0.56) for the RP353 dyke of the TDS (Fig. 3C).

4.3. *Palaeomagnetism*

Palaeomagnetic results from the Hartley Formation are summarized in Table 3. Examples of demagnetization behaviour and summaries of the remanent magnetization components are shown in Fig. 4.

4.3.1. *Hartley Hill Quarry*

Our samples from Hartley Hill quarry overlap with, and extend, the limited sampling originally done by Evans et al. (2002), who only took three oriented samples per flow unit (Fig. 2). We repeated LSQ analyses on specimen data of Evans et al. (2002) and combined it with our own data-set to calculate component means (Table 3).

Random low-coercivity components and a present field-like viscous remanent magnetization were successfully removed in most samples during initial demagnetization steps. An intermediate coercivity component (INT component) was recognized in samples from three of the sites (i.e., two samples from site HAA, as well as all the samples from sites HAF and HAH). This component is generally poorly constrained, but displays shallow inclination and southward declination in present geographic coordinates. High coercivity components (HIG1 component) were removed by demagnetization steps between 35 and 100 mT, either as origin-seeking linear trajectories of demagnetization, or as stable endpoints of demagnetization. These define generally westerly and moderately steep to very steep downwards directed vectors in present geographic coordinates (Figs. 4 and 5C). This component was present in all of the sampled volcanic units at

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Table 1. Summary of SHRIMP geochronological data on zircon.

Grain.Spot	U/Th	^{206}Pbc	^{206}Pb*	^{207}Pb/	±2σ	^{206}Pb/	±2σ	^{207}Pb/	±2σ	^{206}Pb/	±2σ	^{207}Pb/	±2σ	Error	Discordance
		(%)	ppm^2	^{235}U	% err.	^{238}U	% err.	^{235}U		^{238}U (Ma)		^{206}Pb (Ma)		correlation	(%)
							BOS 10								
1.1	4.12	0.11	61.4	12.80	1.5	0.5096	1.3	0.1822	0.66	2655	29	2673	11	0.895	1
2.1	2.57	0.08	104.0	30.07	2.3	0.6183	1.3	0.3528	1.90	3103	32	3719	29	0.556	17
3.1	1.44	0.08	63.0	7.13	1.6	0.3390	1.2	0.1330	0.95	2118	22	2138	17	0.794	1
4.1	1.11	0.41	78.6	6.84	1.4	0.2904	1.2	0.1709	0.78	1644	17	2566	13	0.830	36
5.1	3.93	0.47	94.0	4.30	1.4	0.2430	1.1	0.1283	0.84	1403	14	2075	15	0.800	32
5.1	1.70	0.15	64.7	6.43	1.4	0.3602	1.2	0.1293	0.78	1983	21	2089	14	0.842	5
7.1	1.98	0.20	58.6	7.21	1.5	0.4041	1.3	0.1293	0.75	2188	23	2089	13	0.860	–5
8.1	1.34	1.24	119.0	4.32	1.5	0.2038	1.1	0.1536	1.00	1196	12	2386	17	0.729	50
9.1	3.94	0.17	97.1	6.51	1.3	0.3724	1.2	0.1267	0.64	2041	20	2053	11	0.876	1
10.1	5.91	0.35	114.0	6.73	1.3	0.2992	1.1	0.1631	0.66	1687	17	2488	11	0.862	32
11.1	0.94	0.09	83.8	6.37	1.3	0.3743	1.2	0.1234	0.62	2050	20	2006	11	0.881	–2
12.1	1.54	0.11	61.6	6.61	1.4	0.3803	1.2	0.1261	0.75	2078	22	2044	13	0.856	–2
13.1	1.87	0.04	109.0	22.60	1.3	0.6491	1.2	0.2530	0.41	3225	31	3204	7	0.946	–1
14.1	0.83	0.16	56.5	11.70	1.5	0.4747	1.3	0.1788	0.68	2504	27	2642	11	0.886	5
15.1	2.49	0.14	60.9	13.22	1.5	0.5185	1.3	0.1849	0.69	2693	28	2698	11	0.882	0
16.1	2.61	0.36	84.7	5.46	1.5	0.3092	1.2	0.1281	0.86	1737	19	2072	15	0.818	16
17.1	1.54	0.47	163.0	9.26	1.3	0.3732	1.1	0.1800	0.69	2045	19	2653	11	0.843	23
18.1	1.53	1.54	94.6	2.72	1.8	0.1474	1.1	0.1337	1.40	886	9	2147	24	0.620	59
19.1	1.88	0.37	111.0	6.99	1.3	0.3261	1.1	0.1555	0.63	1819	18	2407	11	0.871	24
20.1	0.48	0.39	78.1	6.67	1.3	0.3040	1.2	0.1591	0.67	1711	17	2446	11	0.864	30
21.1	1.35	0.08	157.0	18.91	1.2	0.5883	1.1	0.2332	0.36	2982	27	3074	6	0.952	3
22.1	2.12	0.27	115.0	23.28	1.5	0.6515	1.4	0.2592	0.57	3234	35	3242	9	0.924	0
23.1	1.95	0.00	75.0	12.60	1.4	0.5147	1.2	0.1776	0.55	2677	27	2630	9	0.914	–2
26.1	2.10	0.13	56.7	13.49	1.5	0.5143	1.4	0.1903	0.71	2675	30	2745	12	0.886	3
27.1	2.39	0.35	122.0	13.11	1.2	0.3819	1.1	0.2490	0.44	2085	20	3178	7	0.931	34
28.1	4.43	2.14	127.0	5.51	2.1	0.3029	1.1	0.1320	1.80	1706	17	2124	32	0.521	20
29.1	0.72	0.55	73.5	7.90	1.6	0.3389	1.3	0.1691	0.98	1881	20	2548	16	0.788	26
30.1	2.78	0.46	76.7	5.40	1.4	0.3041	1.2	0.1287	0.86	1712	18	2080	15	0.806	18
31.1	0.81	1.10	88.8	6.54	1.6	0.2211	1.1	0.2146	1.10	1288	13	2941	18	0.714	56
32.2	1.62	0.48	82.0	5.69	1.4	0.2527	1.1	0.1632	0.81	1452	14	2489	14	0.809	42

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33.2	1.56	0.07	22.6	5.80	1.7	0.3506	1.4	0.1200	10.00	1937	24	1957	18	0.820	1
34.1	19.90	0.35	93.5	4.46	1.6	0.2595	1.1	0.1247	1.10	1487	15	2024	20	0.706	27
35.2	0.94	0.43	45.7	4.71	1.5	0.1908	1.2	0.1700	0.83	1126	12	2557	14	0.823	56
36.2	1.31	0.10	31.7	6.06	1.8	0.3518	1.3	0.1249	1.20	1943	22	2028	22	0.727	4
37.1	1.94	0.44	71.4	4.61	1.5	0.2602	1.2	0.1284	0.89	1491	16	2076	16	0.807	28
38.1	1.63	0.16	71.9	9.78	1.5	0.4497	1.2	0.1577	0.81	2394	24	2431	14	0.833	2
39.2	1.30	0.10	16.2	5.40	1.9	0.3311	1.5	0.1183	1.10	1843	24	1931	20	0.797	5
40.1	1.67	0.07	30.2	12.84	1.6	0.5026	1.4	0.1852	0.71	2625	31	2700	12	0.894	3
41.1	1.72	0.15	54.2	6.38	1.5	0.3725	1.3	0.1241	0.76	2041	22	2017	13	0.857	−1
42.1	2.19	0.00	55.5	14.13	1.8	0.5492	1.4	0.1866	1.20	2822	31	2712	20	0.752	−4
43.1	2.10	0.53	94.6	8.75	1.3	0.3404	1.1	0.1865	0.63	1889	19	2711	10	0.875	30
44.1	3.44	0.24	27.9	6.58	2.0	0.3785	1.5	0.1261	1.30	2070	27	2044	22	0.769	−1
45.1	2.52	0.31	56.6	12.35	1.5	0.5016	1.3	0.1786	0.77	2620	29	2639	13	0.868	1
46.1	2.81	0.22	113.0	10.95	1.2	0.3963	1.1	0.2005	0.48	2152	21	2830	8	0.920	24
47.1	1.45	0.61	73.3	4.86	1.5	0.2711	1.2	0.1300	0.99	1546	16	2098	17	0.764	26
48.1	1.33	0.06	93.7	13.63	1.3	0.5298	1.2	0.1866	0.51	2741	27	2712	9	0.920	−1
49.1	2.38	0.32	137.0	6.71	1.2	0.2927	1.1	0.1662	0.50	1655	16	2520	8	0.905	34
50.1	1.38	0.07	96.1	11.57	1.3	0.4781	1.2	0.1755	0.50	2519	25	2610	8	0.920	3
51.1	2.64		66.9	6.62	1.4	0.3786	1.2	0.1267	0.65	2070	21	2053	12	0.880	−1
52.1	1.90	0.05	64.5	6.49	1.4	0.3757	1.2	0.1253	0.71	2056	22	2032	13	0.865	−1
53.1	1.55	0.06	97.6	20.37	1.4	0.6034	1.3	0.2448	0.43	3044	31	3152	7	0.947	3
54.1	1.58	0.06	86.9	6.85	1.3	0.3855	1.2	0.1288	0.60	2102	21	2082	11	0.889	−1
BH MDK															
1.1	1.77	0.10	29.4	5.62	1.8	0.3452	1.4	0.1181	1.10	1912	23	1928	20	0.776	1
2.1	1.52	0.00	55.9	5.70	1.6	0.3510	1.2	0.1177	1.00	1939	21	1922	19	0.766	−1
3.1	1.33		135.0	5.62	1.2	0.3471	1.1	0.1174	0.42	1921	18	1917	8	0.935	0
4.1	1.47	0.00	24.0	5.43	1.7	0.3325	1.4	0.1185	0.93	1851	23	1933	17	0.834	4
5.1	1.45	0.04	87.4	5.65	1.3	0.3474	1.2	0.1180	0.53	1922	19	1927	9	0.910	0
5.2	1.25	0.06	79.2	5.62	1.3	0.3838	1.2	0.1173	0.47	1905	20	1916	9	0.929	1
6.1	1.66	0.00	36.2	5.66	1.5	0.3431	1.3	0.1196	0.79	1901	22	1949	14	0.858	2
7.1	1.31	0.00	88.9	5.60	1.4	0.3460	1.3	0.1175	0.52	1915	21	1918	9	0.925	0
8.1	1.93	0.00	133.0	5.61	1.2	0.3465	1.1	0.1174	0.43	1918	18	1917	8	0.932	0

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Table 1. (*Contined*).

Grain.Spot	U/Th	^{206}Pbc	^{206}Pb*	^{207}Pb/	±2σ	^{206}Pb/	±2σ	^{207}Pb/	±2σ	^{206}Pb/	±2σ	^{207}Pb/	±2σ	Error	Discordance
		(%)	ppm^2	^{235}U	% err.	^{238}U	% err.	^{235}U		^{238}U (Ma)		^{206}Pb (Ma)		correlation	(%)
9.1	1.54	0.00	41.4	5.68	1.5	0.3473	1.3	0.1185	0.75	1922	22	1934	13	0.866	1
10.1	1.64	0.02	47.8	5.61	1.5	0.3449	1.3	0.1180	0.77	1910	21	1926	14	0.855	1
11.1	1.32	0.05	75.7	5.67	1.3	0.3499	1.2	0.1175	0.50	1934	19	1919	9	0.917	–1
12.1	1.17		78.2	5.71	1.3	0.3510	1.2	0.1180	0.40	1939	21	1925	7	0.950	–1
13.1	1.64	0.00	55.3	5.59	1.2	0.3441	1.2	0.1179	0.47	1906	19	1924	9	0.925	1
14.1	1.77	0.00	31.2	5.54	1.4	0.3420	1.3	0.1175	0.63	1896	21	1918	11	0.895	1
15.1	1.26	0.02	187.0	5.59	1.2	0.3454	1.2	0.1174	0.27	1912	19	1917	5	0.973	0
16.1	1.44	0.01	79.5	5.26	1.2	0.3234	1.1	0.1179	0.42	1806	18	1924	8	0.936	6
17.1	1.42	0.05	45.5	5.72	1.3	0.3536	1.2	0.1174	0.57	1952	20	1917	10	0.905	–2

Notes: Pbc is common Pb.
Pb* is radiogenic Pb.

Hartley Hill, but its resolution progressively deteriorated towards the top of the quarry perhaps as an effect of increased weathering. The HIG1 component in the unit HAE (α_{95} = 43.7° and k = 4.0) and in unit HAH (where it was only recorded by two samples) were not included in the component mean calculation. In other upper units from the quarry (i.e., HL04, HAF and HAG), the HIG1 component was present in 30–50% of samples, and α_{95} values range from 21.9° to 23.4°. A cut-off of $\alpha_{95} \leq 16°$ for inclusion of values in mean calculations is usual, but a higher cut-off of 25° was employed here. The HIG1 component in unit HL04 is distinct from the other sites (Fig. 5C), in that it is shallow and upwards directed, and was excluded from the mean HIG1 component calculation (Table 3).

4.3.2. O'Doneghue Farm

At O'Doneghue Farm, lava of the Hartley Formation is foliated and appears to be more altered (i.e., more greenish). Random lower coercivity components are removed in all specimens with demagnetization below 60–80 mT. At higher levels of demagnetization a consistent east-south-easterly and upward directed (in geographic coordinates) characteristic component was revealed, and is labelled HIG2 (Fig. 5D). The easterly and upward directed grouping of high coercivity components restores to a north-easterly and downward directed position upon structural correction. This component (viewed either in geographic or tilt-corrected coordinates) has not been identified at Hartley Hill.

4.3.3. Pramberg Farm

Quartz porphyry outcrops as low hills towards the southeast of Volop Group quartzite ridges on the Pramberg Farm. The quartzite dips steeply (at ~53°) towards the west, which is taken as the attitude of the porphyry as well. Demagnetization of nine specimens yielded disappointingly disparate results that were challenging to interpret. Apart from a couple present field-like components, a easterly upwards directed component (in geographic coordinates) emerges in three specimens, which is comparable to the HIG2 component identified at O'Doneghue (Fig. 5D). At both Pramberg and O'Doneghue, the HIG2 component is very poorly constrained and is excluded from further discussion.

5. Discussion

5.1. Age of the Hartley Formation volcanism and related magmatism

The dates obtained on zircons from the quartz porphyry in drillcore from Mooihoek (MHK1/86) demonstrate an age of 1920 ± 4 Ma, which is within error of a new SIMS age of 1915 ± 3 Ma from the Pramberg porphyry (Dave Cornell, pers. comm. 2015). Variability in ages is unsurprising for rift related volcanism, as often isolated graben systems each recorded a unique sedimentary and volcanic fill (e.g., de Kock et al. 2012). The TDS sample RP353 was here dated to 1923 ± 6 Ma, which is comparable to ca. 1927 Ma ages obtained for broadly coeval mafic sills in Botswana by Hanson et al. (2004). These ages are also comparable with those of the youngest concordant detrital zircon grains (1931 ± 20 and 1957 ± 18 Ma), from sample BOS10, which are considered to

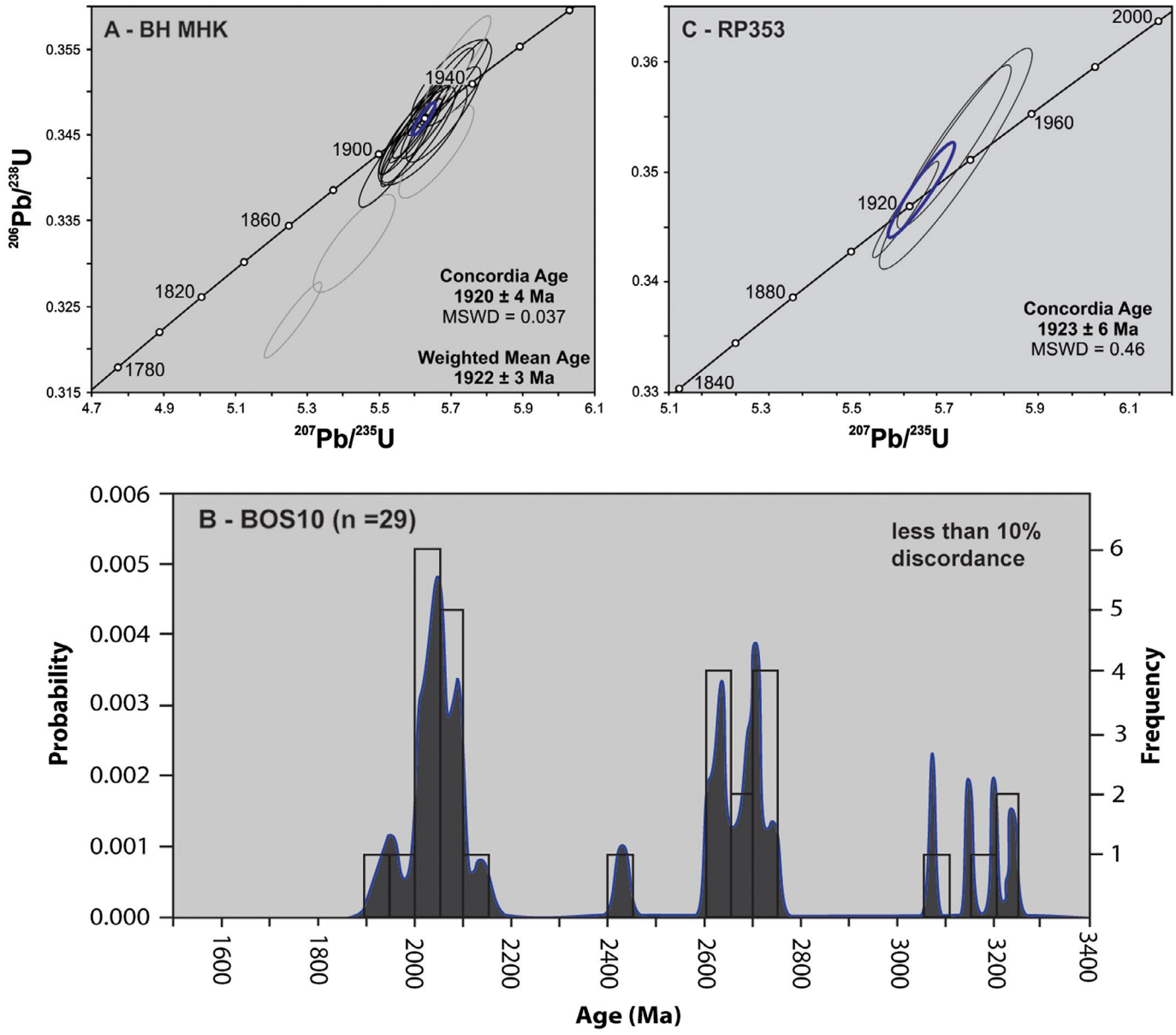


Fig. 3. U–Pb geochronological results from SHRIMP on zircon and ID-TIMS on baddeleyite. **A**. Concordia diagram illustrating ages of zircon grains from the Mooihoek quartz prophyry (sample BH-MHK) of the Hartley Formation. **B**. Spectrum of detrital populations of sample BOS10 from the Boegoeberg Dam Formation on the farm Bovenseekoebaard shown as an age histogram and as a relative probability curve. Included are $^{207}Pb/^{206}Pb$ ages of individual zircon grains that were less than 10% discordant. **C**. Concordia diagram for the analyses of baddeleyite fractions from a dyke of the Tsineng Dyke Swarm in drill core from borehole RP353 at Smartt Mine in the Kalahari Manganese Field near Tsineng.

Table 2. Summary of ID-TIMS geochronological data on baddeleyite.

Analysis No. (number of grains)	U/ Th	Pbc/ Pbtot[1]	$^{206}Pb/^{204}Pb$ raw[2]	$^{207}Pb/^{235}U$	±2σ % err.	$^{206}Pb/^{238}U$ [corr.][3]	±2σ % err.	$^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$	$^{207}Pb/^{206}Pb$ [age, Ma]	±2σ	Concordance
Bd-1 (9)	4.1	0.384	121.5	5.7238	1.78	0.35232	1.73	1935.0	1945.6	1923.5	11.2	1.011
Bd-2 (12)	5.8	0.213	286.3	5.7280	2.41	0.35118	2.35	1935.6	1940.2	1930.6	14.2	1.005
Bd-3 (16)	4.6	0.075	828.4	5.6202	1.04	0.34661	1.03	1919.2	1918.4	1920.0	5.2	0.999

[1]Pbc = common Pb; Pbtot = total Pb (radiogenic + blank + initial).
[2]Measured ratio, corrected for fractionation and spike.
[3]Isotopic ratios corrected for fractionation (0.1% per amu for Pb), spike contribution, blank (1 pg Pb and 0.1 pg U) and initial common Pb. Initial common Pb corrected with isotopic compositions from the model of Stacey and Kramers (1975) at the age of the sample.

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Table 3. Summary of palaeomagnetic results.

Location (Strike/Dip)	Site	*n*/*N*	Declination (in °)	Inclination (in °)	α_{95} (in °)	*k*	Latitude (in °N)	Longitude (in °E)	(dp, dm) or A95 (in °)
			Present field-like (geographic coordinates)						
Hartley Hill (89°/22°S)	HAA and HL01*	9/16	332.4	−52.4	7.4	48.9			
(60°/7°SE)	HAB and HL02*	10/15	50.2	−63.8	9.8	25.2			
(32°/7°E)	*HAD*	*4/5*	*43.0*	*−65.4*	*40.4*	*4.2*			
(90°/22°S)	HAC and HL03*	9/11	15.6	−64.3	17.2	9.9			
(120°/4°SW)	*HL04**	*3/4*	*346.4*	*−46.9*	*37.4*	*7.9*	No poles calculated for present present field-like Component A		
(90°/22°S)	HAE and HL05*	12/14	348.4	−80.0	16	8.4			
(88°/22°S)	HAF and HL06*	5/14	40.9	−28.3	21.6	13.5			
(66°/10°SE)	HAG and HL07*	4/14	18.5	−43.5	18.5	25.6			
Pramberg (107°/53°W)	HBA	3/9	349.3	−61.2	14.2	51.2			
	A Mean	7/9	14.8	−59.2	17.4	13			
			Component INT (geographic coordinates)						
Hartley Hill (88°/22°S)	HAF and HL06*	7/7	199.7	−16.8	9.8	39.2	48.8	53.2	(5.2, 10.1)
(75°/6°SE)	*HAH* and *HL08**	8/12	154.3	7.8	30.5	4.3	55.7	332.6	(15.5, 30.7)
	JP-11**	5/10	180.5	−28.1	5.8	69.2	50.1	26.1	(3.5, 6.4)
			Component HIG1						
Location (strike/dip)	Site	*n*/*N*	Declination and inclination (in °) (geographic coordinates)	Declination and Inclination (in °) (tilt-corrected coordinates)	α_{95} (in °)	*k*	Latitude (in °N)	Longitude (in °E)	(dp, dm) or A95 (in °)
Hartley Hill (89°/22°S)	HAA and HL01*	8/16	293.4, 28.0	289.1, 26.6	13.0	19.2	9.6	314.4	(7.7, 14.1)
(60°/7°SE)	HAB and HL02*	8/15	305.4, 35.0	304.9, 36.9	14.1	16.3	18.0	329.0	(9.7, 16.5)
(90°/22°S)	HAC and HL03*	3/11	318.4, 44.8	322.8, 50.6	18.6	30.1	21.0	349.2	(16.8, 25.0)
(32°/7°E)	HAD	3/5	291.7, 31.3	291.2, 34.6	15.5	42.6	8.6	319.8	(10.2, 17.8)
(90°/22°S)	*HAE* and *HL05**	*5/14*	*310.0, 26.6*	*310.9, 25.1*	*43.7*	*4.0*	*27.2*	*327.0*	(*25.3, 47.0*)
(88°/22°S)	HAF	4/14	307.5, 36.1	306.4, 38.0	23.4	12.3	18.6	330.5	(16.3, 27.7)
(66°/10°SE)	HAG and HL07*	4/14	312.0, 40.7	312.2, 44.1	22.7	17.3	19.3	337.9	(17.8, 28.4)
(75°/6°SE)	HAH and HL08*	2/12	303.7, 36.1	302.7, 37.0	–	–	16.4	327.6	–
(120°/4°SW)	*HL04**	*4/4*	*274.9,-16.9*	*275.7, −15.2*	*21.9*	*13.9*	*6.9*	*257.7*	(*11.5, 22.5*)

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	Hartley Mean	6/9		303.3, 39.0	10.6	40.6	16.1	329.9	11.0	
Moshaneng (*no correction*)	JP-9**	14/0	324.1, 7.2	324.1, 7.2	6.3	40.2	45	329.4	(3.2, 6.4)	
(*no correction*)	JP-10**	13/1	315.0, 0.9	315.0, 0.9	7.5	31.6	39.6	318.5	(3.7, 7.5)	
Hartley Hill + Moshaneng	Hartley Mean	8/12		308.2, 30.6	14.6	15.4	22.7	328.6	11.7	
Component HIG2 (geographic coordinates)										
Pramberg (107°/53°W)	HBA	4/9	114	−66.4	27.5	9.1	16.5	345.3	(28.9, 39.3)	
O'Doneghue (99°/62°SW)	HCA	5/8	120	−55.5	26	7.7	4.7	338.1	(26.5, 37.1)	

Notes: *Italic font* = data excluded from mean calculation.
n/*N* = number of samples included in component site mean/total number of samples collected at site.
*Sampled by Evans et al. (2002).
**Sampled by Hanson et al. (2004).

represent the timing of eruption of the Boegoeberg Dam mafic lava unit. This unit may thus be considered the age equivalent to the Hartley Formation as suggested by Cornell et al. (1998). An age range of 1927–1915 Ma can thus be placed on the Hartley Formation bimodal igneous rocks from the work in this study combined with the work of Cornell et al. (1998) and Hanson et al. (2004).

5.2. *Identifying the potential Hartley LIP and dynamics of late Palaeoproterozoic rifting*

With the now well-dated 1.93–1.91 Ga magmatic units of the Hartley Formation, Boegoeberg Dam Formation, TDS, Moshaneng sills and Trompsburg Complex, a large scale bimodal magmatic event is be recognized. We propose that these magmatic units be incorporated into a LIP, the Hartley Event. Its size (footprint on the order 4×10^5 km^2) is compatible with the magnitude of known LIP dimensions (Coffin & Eldholm 1993; Ernst 2014), but the duration of emplacement for the Hartley igneous rocks is longer than most typical LIPs according to the LIP definition (Ernst 2014).

Tinker et al. (2002) reported on a seismic reflection profile spanning the western margin of the northern Kaapvaal Craton (see Fig. 6 for the exact locality), and presented insight into the dynamics and structure associated with the extensional event that culminated in the emplacement of the Hartley LIP. Late Palaeoproterozoic extension was initiated before, but continued during eruption of the Hartley volcanic rocks (Tinker et al. 2002). According to Tinker et al. (2002), an unknown amount of extension occurred along such deep crustal structures as the Moshaweng Normal Fault (Fig. 6.), a prominent N–S trending normal fault that was reactivated or initiated by late Palaeoproterozoic rifting. According to Moen (2006), early graben formation is not evident from surface exposures, but the predominance of rudites in drillcore north of Olifantshoek suggest that the basal parts of the Elim Group were deposited in grabens. Late Palaeoproterozoic deformation along structures like the Moshaweng Fault was associated with the development of a series of normal faults, and caused relative uplift of the Elim Group in the form of a rollover anticline (Tinker et al. 2002). Continued extension resulted in the formation of half-grabens bounded on their east side by structures like the Moshaweng Fault and an anticlinal structure to their west side (Fig. 6 inset). The half-grabens were then later filled by erosional products, such as the Elim Group (i.e., the conglomerates and quartzites at the base of the Hartley Formation), followed by volcanic products of the Hartley Formation. A system of similar half-graben structures probably punctuated by several volcanic centres extended towards the south along the western Kaapvaal margin, as evidenced by the ca. 1.93 Ga Boegoeberg Dam Formation (Fig. 6). Flexural analyses (Tinker et al. 2004) have shown that the elastic lithosphere along this western margin was significantly reduced (to as little as 7.5 km, compared to the present day 60–70 km thickness) between ca. 1.93 and 1.75 Ga, suggesting that a significant amount of crustal thinning took place during this extensional event.

Our 1923 ± 6 Ma age for the TDS, which is truncated by the Kheis Province and extends towards the east-north-east, with a strike length of ~500 km into the cratonic hinterland (Fig. 6), probably dates an arm of NW-SE extension associated with rift-

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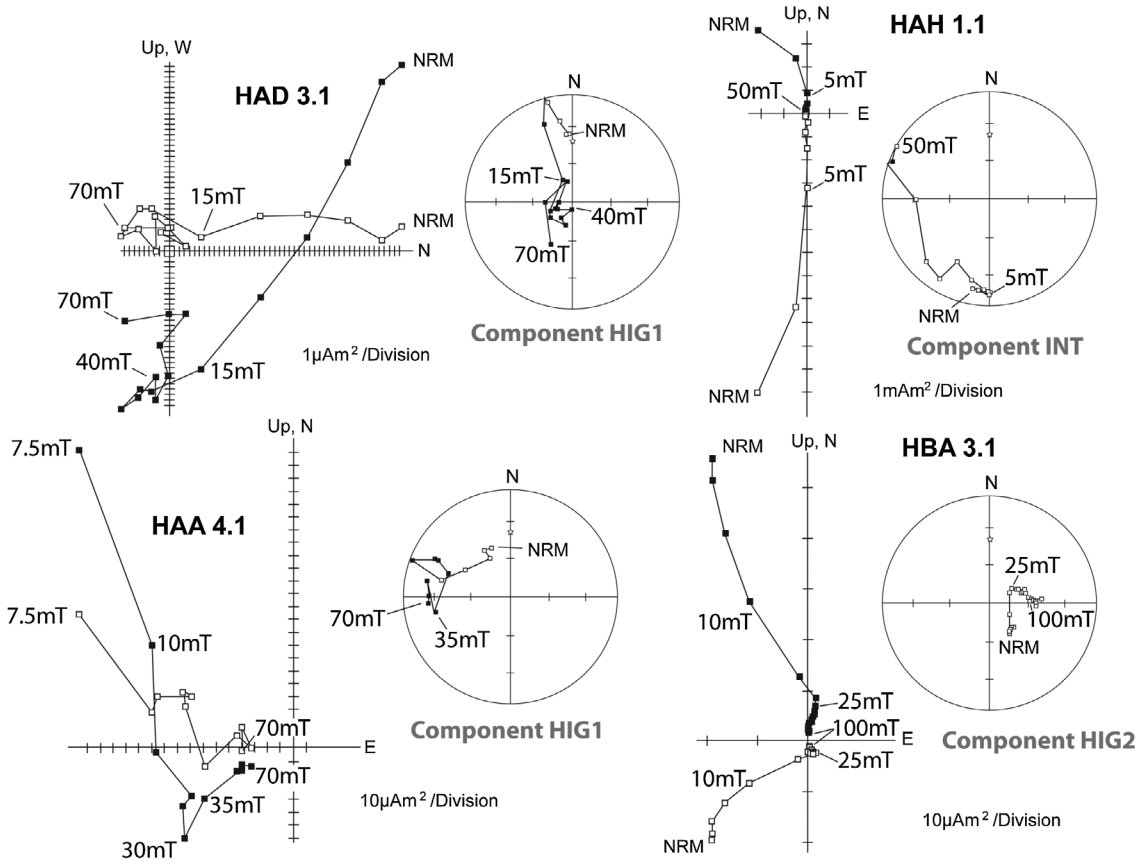


Fig. 4. Representative demagnetization behaviour for the Hartley volcanic rocks in geographic coordinates. Open symbols plot in the upper hemisphere of equal area plots, while solid symbols plot in the lower hemisphere. Open symbols on the orthogonal plots represent the projection of vector endpoints onto the horizontal plane, while closed symbols represents the projection of vector endpoints onto a vertical plane.

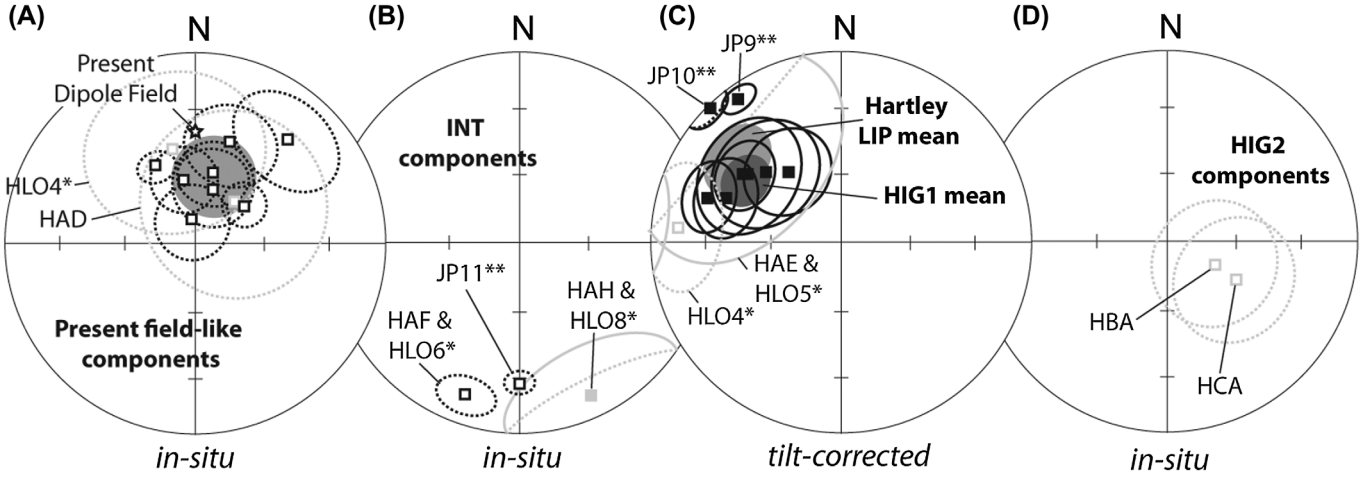


Fig. 5. Site and component means for magnetic components identified during this study. Ellipses are cones of 95% confidence (or α_{95}) about component means (shown as squares). Stippled ellipses and open squares plot in the upper hemisphere of the equal area diagrams, while solid ellipses and closed squares are in the lower hemisphere. Lighter shaded symbols and ellipses are from sites that were excluded from component mean calculations. Data from Evans et al. (2002) are identified by an *, whereas those from Hanson et al. (2004) by **. **A**. Low coercivity, present Earth field-like components (in geographic coordinates) with the present dipole field of the Earth shown as a star. **B**. Component INT site means compared to a similar component from a sill in Botswana (JP11), in geographic coordinates. **C**. Component HIG1 site means in tilt-corrected coordinates compared to similar components from sills in Botswana. **D**. Poorly constrained Component HIG2 site means in geographic coordinates.

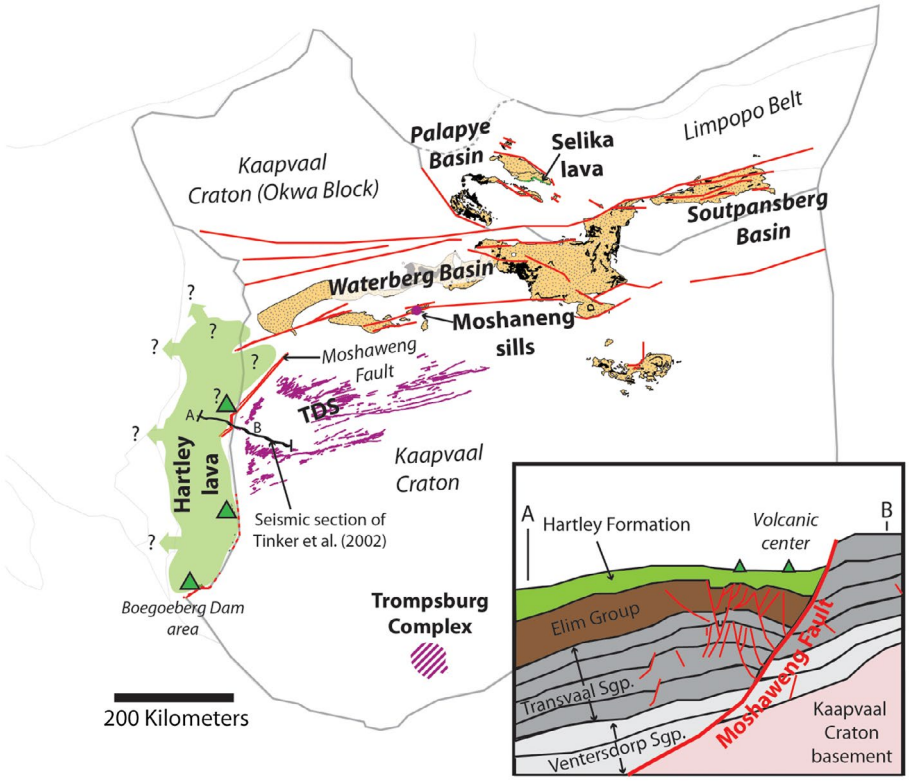


Fig. 6. Possible original extent and structural setting of the Hartley LIP event in relation to areas of red bed deposition across the Kaapvaal Craton and dominant volcanic centers (green triangles). Inset displays half-graben and rollover anticline structures west of the Moshaweng Fault (after Tinker et al. 2002).

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ing. The Moshaweng Fault can be traced further towards the north where its trend swings towards the ENE–WSW (Fig. 6), becoming parallel to the trend of the TDS. Goldberg (2010) interpreted the TDS as part of a failed arm of the Palaeoproterozoic rifting along the western Kaapvaal Craton margin. In addition, the present day outcrop distribution of the Waterberg and Soutpansberg basins occur along a similar ENE–WSW axis as that of the TDS, and these basins have been proposed to occupy an aulocogen (Cornell 1987) or series of rifts (e.g., Barker 1983), and fault-bounded pull-apart basins (Eriksson et al. 2006), that acted as a long-lived (2.045 Ga to at least 1.83 Ga) areas of sediment accumulation.

The emplacement of the 1915 Ma Trompsburg Complex and ca. 1927 Ma Moshaneng Sills in the cratonic interior (Fig. 6), provide evidence for a larger volume of magma generation possibly associated with Palaeoproterozoic rifting. The exact relationship of the Trompsburg Complex to the Hartley LIP and rifting is not fully understood, although it is broadly coeval.

5.3. Hartley Formation palaeopole and the late Palaeoproterozoic APWP of the Kaapvaal Craton

The HIG1 component is a consistent characteristic remanent direction (Fig. 5C) for the Hartley Formation. A palaeopole at 16.1°N, 329.8°E with A_{95} = 11.0° can be calculated by combining six of our site virtual geomagnetic poles (VGPs) (Table 3, Fig. 7B, C). Like Evans et al. (2002), we too cannot be sure that our data effectively average out palaeosecular variation of the Earth's magnetic field. The averaging of secular variation can perhaps be addressed to some extent by adding the results from the ca. 1927 Ma Moshaneng sills (Hanson et al. 2004) to our data for the calculation of a 1927–1915 Ma mean palaeopole (Table 3). VGPs from the ca. 1927 Ma Moshaneng sills plot removed from, but in a generally similar region as our calculated site VGPs for the Hartley Formation. This might be expected given a possible age difference between the Moshaneng Sills and Hartley lava, but could also be due to unrecognized vertical axis rotation of about 15° during the development of the Kheis Province. The 1927–1915 Ma mean palaeopole is located at 22.7°N, 328.6°E with A_{95} = 11.7° (Fig. 7B, C), and rates as a 4 out of 7 on the Q-scale of reliability of Van der Voo (1990). By no means ideal (too few sample sites, lack of a field test and magnetic reversals), our new 1927–1915 Ga Hartley LIP palaeopole can still be considered a significant improvement on the existing VGP of Evans et al. (2002), given the significant increase in the number of samples per cooling unit, and the statistically sound palaeopole calculation.

The identified INT component yields a VGP at 48.8°N and 53.2°E (dp = 5.2, dm = 10.1) for site HAF, where it was best developed. Compared to palaeopoles from the Kaapvaal Craton (Table 4 and Fig. 7A), this VGP is very similar to the 1.2–1.1 Ga poles of the Kaapvaal Craton. A similar magnetic component was identified by Hanson et al. (2004) in a 1927 ± 1 Ma Moshaneng dolerite sill from Botswana (i.e., their JP11 sill, shown for comparison in Fig. 5B), where it was interpreted to be a 1.11 Ga magnetic overprint.

Several new palaeopoles and new age constraints (listed in Table 4) have been published since the last attempted reconstruc-

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Table 4. Poles calculated during this study and a selection of published Late Palaeoproterozoic and Mesoproterozoic poles from the Kaapvaal Craton.

Rock Unit	Abbreviation	Age (Ma)	Age reference	Latitude	Longitude	(dp, dm) or A95	Palaeopole reference
				(in °N)	(in °E)	(in °)	
Palaeoproterozoic palaeoples							
Ongeluk Lava	ONG	2222 ± 13	Cornell et al. (1996)	22	−0.5	5.3	Evans et al. (1997)
Basal Gamagara/Mapedi Formation	BGM	~2200 or	Evans et al. (2002)	2.2	81.9	(7.2, 11.5)	Evans et al. (2002)
Elim Group		~2000	Da Silva (2011)				
Phalaborwa Complex	PC	2061 ± 1	Reischmann (1995)	27.7	35.8	6.6	Letts et al. (2010)
Bushveld Complex	BC	2054 ± 1	Scoates & Friedman (2008)	19.2	30.8	5.8	Letts et al. (2009)
Bushveld B1 sills	B1	2058 ± 6	Wabo et al. (2015a)	13.1	44	14.3	Wabo et al. (2015a)
Uitkomst Complex	UC	2055 ± 7	Wabo et al. (2015b)	28.7	58.5	(6.2, 9.4)	Wabo et al. (2015b)
Waterberg unconformity bounded sequence 1	WUBS1	≤2054 ± 4	Dorland et al. (2006)	36.5	51.3	10.9	de Kock et al. (2006)
Vredefort Impact	VRED	2023 ± 4.0	Kamo et al. (1996)	21.8	44.5	(11.3, 15.4)	Carpozen et al. (2006)
Witwatersrand Supergroup Overprint	WITS	1945–2050	Layer et al. (1988)	26.1	22.3	(7.9, 10.3)	Layer et al. (1988)
Waterberg unconformity bounded sequence 2	WUBS2	>1930	de Kock et al. (2006)	−10.5	330.4	9.8	de Kock et al. (2006)
Hartley Formation VGP	HAR VGP	1928 ± 4	Cornell et al. (1998)	12.5	332.8	16	Evans et al. (2002)
Hartley Formation Palaeopole	HIG1	1921 ± 4	THIS STUDY	16.1	329.9	11	THIS STUDY
JP9 Moshaneng Sill	JP9	1928 ± 1	Hanson et al. (2004)	45	329.4	(3.2, 6.4)	Hanson et al. (2004)
JP10 Moshaneng Sill	JP10	1928 ± 1	Hanson et al. (2004)	39.6	318.5	(3.7, 7.5)	Hanson et al. (2004)
Hartley LIP	HAR LIP	1927–1915	THIS STUDY	22.7	328.6	11.7	THIS STUDY recalculated using Evans et al. (2002)
Sand River Dykes	SRD	1876 ± 73	Morgan and Briden (1981)	2.3	9.1	10.3	Morgan and Briden (1981)
Post-Waterberg Dolerites	PWD	1879–1872	Hanson et al. (2004)	15.6	17.1	8.9	Hanson et al. (2004)
Post-Phalaborwa Dykes	PPD	1857–1839	Olsson et al. (2015)	7.6	12.1	11.8	Letts et al. (2010)

(*Continued*)

Table 4. (*Continued*).

Rock Unit	Abbreviation	Age (Ma)	Age reference	Latitude	Longitude	(dp, dm) or A95	Palaeopole reference
				(in °N)	(in °E)	(in °)	
Post-Bushveld Dykes	PBD	1857–1839	Olsson et al. (2015)	12.6	24.1	10.8	Letts et al. (2010)
Black Hills Dyke Swarm	BHD	1857–1840	Olsson et al. (2015)	9.4	352	(4.3, 5.8)	Lubnina et al. (2010)
Mashonaland Sills (Zimbabwe)	MASH	1886–1872	Hanson et al. (2011) Söderlund et al. (2010)	6.5	338.5	5	Hanson et al. (2011)
			Mesoproterozoic Palaeoploes				
Premier Kimberlite	P	~1200–1180	Hanson et al. (2006)	51.3	37.9	(4.0, 7.5)	Hargraves & Onstott (1980)
Kalkpunt Formation	KP	1093 ± 7	Pettersson et al. (2007)	57	183	(4.0, 7.0)	Briden et al. (1979)
Umkondo Dolerites	UMK	1108–1112	de Kock et al. (2014)	64	38.8	3.7	Gose et al. (2006)
Namaqua Eastern Zone	NEZ	<1165	Pettersson et al. (2007)	44.9	21.5	12.8	Evans et al. (2002)
Namaqua-Natal	NAM	~1000	Gose et al. (2013)	9.4	329.5	17.8	Gose et al. (2013)
Metamorphic Rocks							
Hartley Component INT	INT	~1100	THIS STUDY	48.8	53.2	(5.2, 10.1)	THIS STUDY

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tion of the Palaeoproterozoic APW path of the Kaapvaal Craton (de Kock et al. 2006). In the proposed path of de Kock et al. (2006), a pre-Bushveld age for the Elim Group was assumed following the work of Evans et al. (2002). Recent indications by Da Silva (2011) suggest that the Elim Group may post-date emplacement of Bushveld Complex as proposed by Moen (2006), which may have significant implications for how the APW path is constructed, and paths for both age options of the Elim Group are presented (Fig. 7).

5.3.1. A pre-Bushveld age for the Elim Group

In the pre-Bushveld age option for the Elim Group, the late Palaeoproterozoic APW path of the Kaapvaal Craton (Fig. 7B) begins from the Gamagara-Mapedi Formation palaeopole of Evans et al. (2002) from the base of the Elim Group. From here the path is constrained by a 2060–2024 Ma cluster of palaeopoles (see Table 4 for details on individual poles and their age constraints). This cluster includes poles from the 2060 Ma Phalaborwa Complex (Letts et al. 2010), 2059–2054 Ma Bushveld LIP (Letts et al. 2009; Wabo et al. 2015a, b), the Bushveld Complex-induced magnetic overprint pole from the Mesoarchean Witwatersrand Supergroup (Layer et al. 1988), the younger than 2054 Ma first unconformity-bounded sequence of the Waterberg Group palaeopole or WUBS1 pole (de Kock et al. 2006), and the 2023 Ma Vredefort impact structure (Carpozen et al. 2006). From here the pre-1.9 Ga palaeopole from the second unconformity-bounded Waterberg Group sequence or WUBS2 palaeopole (de Kock et al. 2006) suggests a long southwesterly swath of the APW path (Fig. 7B). Our 1927–1915 Ma mean palaeopole from the Hartley LIP is located at a hairpin of the APW path (Fig. 7B), where the path loops back on itself towards ca. 1.8 Ga poles. The ca. 1.8 Ga cluster includes a palaeopole from the 1.88–1.84 Ga Post-Waterberg sills by Hanson et al. (2004), and palaeopoles from the NNE–SSW and NE–SW trending dykes of the Black Hills Dyke Swarm (Morgan 1985; Letts et al. 2010; Lubnina et al. 2010; Olsson et al. 2015). The Black Hills Dyke Swarm includes the so-called Sand River dykes and post-Bushveld dykes.

5.3.2. A post-Bushveld age for the Elim Group

In this option (Fig. 7C), the APW path starts with the 2.06–2.02 Ga palaeopole cluster as defined above (Section 5.3.1.), then moves a short distance to the ca. 2.0 Ga Elim Group Gamagara–Mapedi Formation palaeopole (Evans et al. 2002). Evans et al. (2002) also illustrated this possibility, but not as their preferred choice. The next youngest palaeopole is the WUBS2 palaeopole (de Kock et al. 2006), but instead of requiring a long southwesterly swath of the path as in Fig. 7B, by choosing an opposite polarity option for this pole, the APW path continues towards the east (Fig. 7C). Palaeopoles can be represented either as a North Pole or South Pole when defining APW paths, given the ambiguity of magnetic reversals in the past. However, the next younger palaeopole in a path is usually placed in the polarity option that minimizes the path length between successive poles. From the

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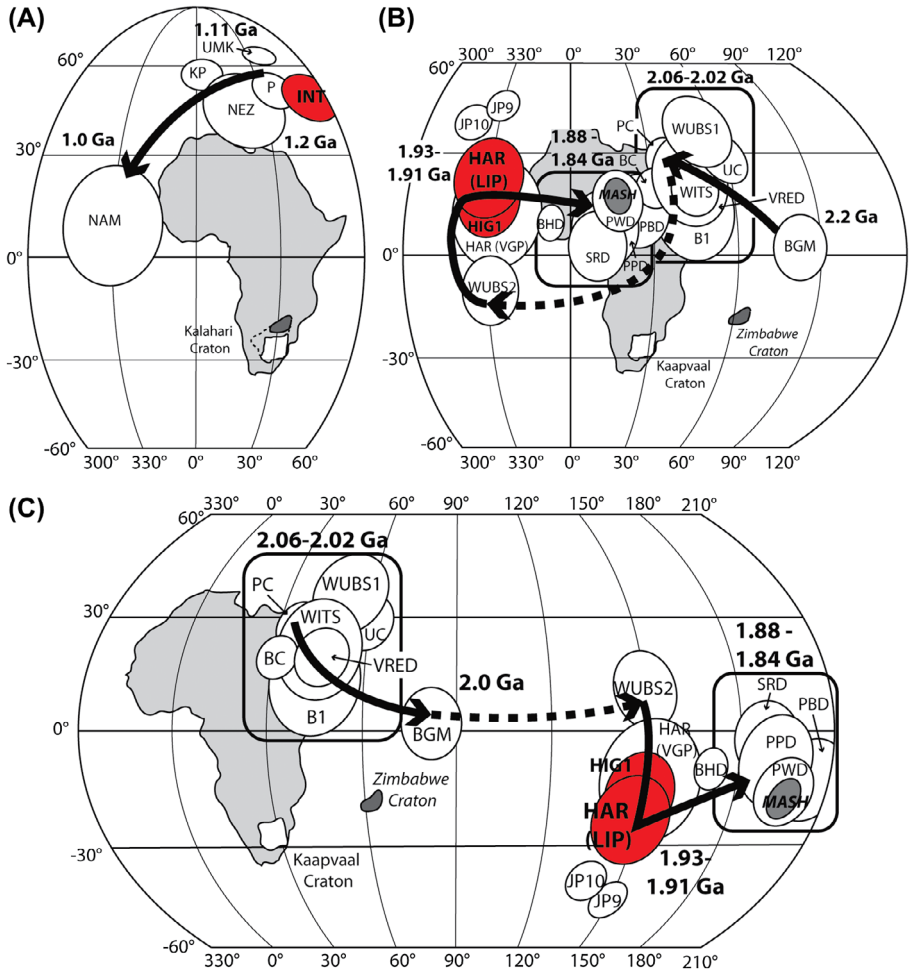


Fig. 7. **A**. Component INT this study (in red) compared with Mesoproterozoic poles from the Kaapvaal Craton. The 1.2–1.0 Ga segment of the Mesoproterozoic APW path is shown as a thick black arrow after Gose et al. (2013). For pole abbreviations and more details on their ages, see Table 3. **B**. Late Palaeoproterozoic APW path (thick black arrow) of the Kaapvaal Craton with a pre-Bushveld age for the Elim Group. **C**. Alternative late Palaeoproterozoic APW path (thick black arrow) for the Kaapvaal Craton with a post-Bushveld age for the Elim Group. The relative position of the Zimbabwe Craton with respect to the Kaapvaal Craton at 1.88 Ga (rotated –39° around an Euler pole at 69.4°S and 50.9°E; Hanson et al. 2004), is shown in both Late Palaeoproterozoic APW path options. The Zimbabwean Mashonaland sills pole (MASH) is shaded, while poles presented in this study are shown in red. APW path is stippled for times of poor data coverage.

WUBS2 palaeopole, the path swings slightly south towards our 1.93–1.91 Ma Hartley LIP palaeopole (opposite polarity option), and continues eastwards towards the opposite polarity option of the 1.88–1.84 Ga pole cluster (Fig. 7C). This radically different interpretation of the Kaapvaal Craton's late Palaeoproterozoic APW path segment eliminates the need for a loop in the APW path of the Kaapvaal Craton.

5.4. The late Palaeoproterozoic magmatic record of the Kaapvaal Craton and constraints on Kalahari Craton assembly

Within basal sections of the Waterberg Group, which is otherwise devoid of volcanic rocks, bimodal volcanic rocks are preserved (yielding 2054 ± 4 and 2021 ± 5 Ma ages; Dorland et al., 2006). The 1927–1915 Ma Hartley LIP (this study) represents the next record of volcanic and magmatic activity for the Kaapvaal Craton. This event may also include the undated mafic volcanic rocks of the lower Selika Formation of the Palapye Group of Botswana. It is followed by extensive 1.88–1.83 Ga intraplate magmatism. Sills that intrude the Waterberg Group (Fig. 1) were dated in the 1879–1872 Ma age range (Hanson et al. 2004), while the extensive NE–SW trending Black Hills Dyke Swarm (Fig. 1) have been shown to have ages ranging from 1857–1839 Ma (Olsson et al. 2015). The post-Waterberg dolerite sills were correlated with the basal volcanic units preserved in the Soutpansberg Group. However, recent dates by Geng et al. (2014) suggest that the Sibasa Formation of the Soutpansberg Group is younger. The 1831 ± 15 and 1832 ± 9 Ma detrital zircon ages from Sibasa Formation by Geng et al. (2014) are closer to the younger range of dolerite dyke ages found by Olsson et al. (2015) for the Black Hills Dyke Swarm. The Kaapvaal Craton likely experienced an extended period of intraplate magmatism starting with the 1879–1872 Ma post-Waterberg sills and ending with the extrusion of the Sibasa Formation at 1832 Ma. Younger, but less-well dated magmatic units are also found in the upper Soutpansberg Group, as well as within the Brulpan Group along the craton's western margin. The age of Palaeoproterozoic

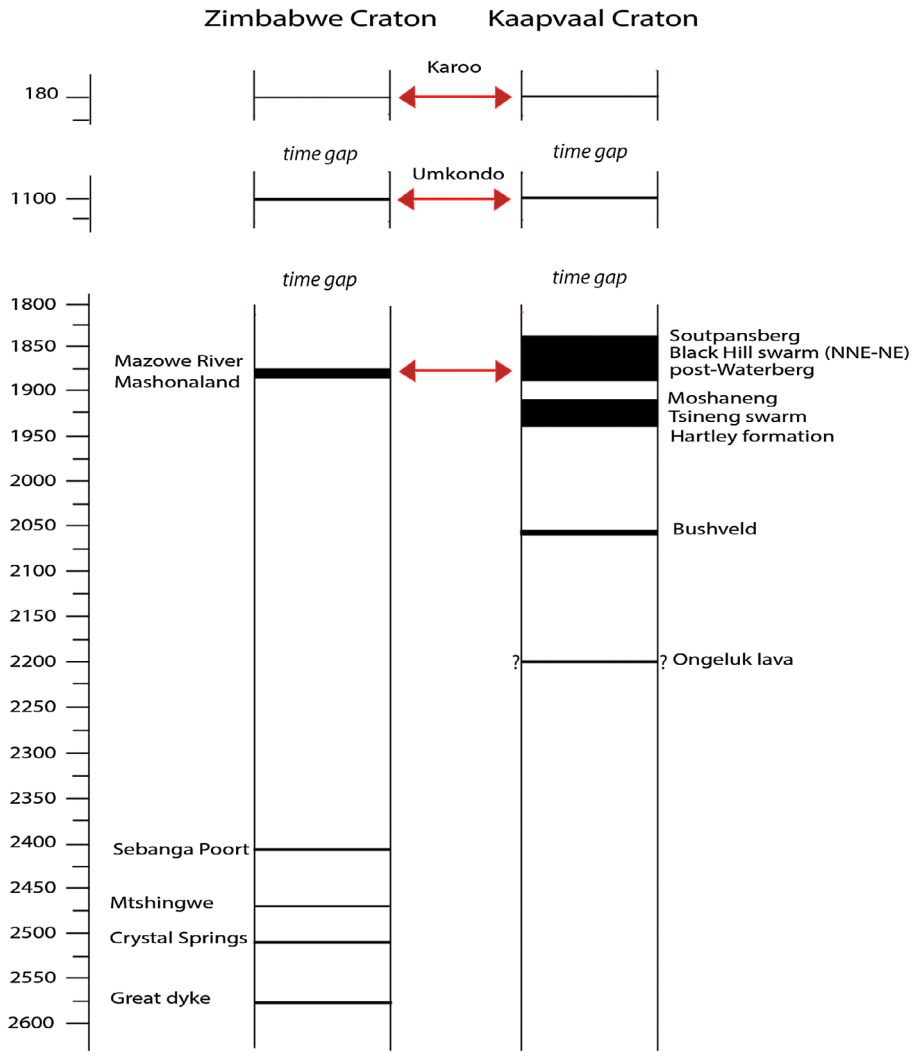


Fig. 8. Magmatic barcodes for the Kaapvaal and Zimbabwe cratons modified after Söderlund et al. (2010).

sills in the Soutpansberg Group are only constrained by a Rb–Sr isochron age of 1797 ± 104 Ma (Barton 1979; recalculated using λ^{87}Rb of Rotenberg et al. 2012). The Brulpan Group (i.e., the Skerpioenpunt Member) yielded a recalculated Rb–Sr whole rock errorchron age of 1095 ± 144 Ma (Barton & Burger 1983; Rotenberg et al. 2012), suggesting Sr-isotopic homogenization during the Namaqua–Natal Orogeny (Barton & Burger 1983). These volcanic rocks might be the equivalents of unaltered dolerite sills and dykes that intrude the Olifantshoek Supergroup within numerous borehole cores. One such dolerite sill crosscuts foliated Hartley Formation south of Olifantshoek on the farm Mamatlun (Fig. 1), and Cornell et al. (1998) reported a Rb–Sr biotite mineral age of 1797 ± 60 Ma (also adjusted using new decay constant value by Rotenberg et al. 2012). A possibility that this younger ca. 1.80 Ga igneous event may be more widespread so as to include the Skerpioenspunt volcanic rocks of the Groblerhoop Group and the Soutpansberg's Ngwanedzi volcanic rocks and sills was highlighted by Dorland et al. (2006), but remains untested.

It has been pointed out by Söderlund et al. (2010) that the 1927–1915 Ma magmatic event appears to be unique for the Kaapvaal Craton, which is in contrast to the 1.88–1.83 Ga event that can be identified in both the Kaapvaal and Zimbabwe cratons. In Zimbabwe, the intrusion of the Mashonaland sills has been dated to between 1886 and 1871 Ma (Söderlund et al. 2010; Hanson et al. 2011). This indicates that the two cratons might have been "nearest neighbours" as defined by Bleeker & Ernst (2006) at ca. 1.88 Ga. However, Hanson et al. (2011) argued for a greater than 2000 km displacement between the Kaapvaal and Zimbabwe cratons based on the difference between their ca. 1.88 Ga palaeopoles (Fig. 7B, C). The Zimbabwe Craton does not share a record of the Kaapvaal Craton Hartley LIP at 1.93–1.91 Ga (Fig. 8), which appears to have been emplaced during an early phase of Kalahari Craton assembly.

6. Conclusion

A SHRIMP U–Pb zircon age of 1920 ± 4 Ma for the Hartley Formation (Olifantshoek Supergroup) on the western margin of the Kaapvaal Craton of South Africa is presented. Together with a new maximum concordant detrital zircon age of 1931 ± 20 Ma

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for a coeval unit 140 km to the south, and a new 1923 ± 6 Ma U–Pb ID-TIMS baddeleyite date from the Hartley-related Tsineng Dyke Swarm, a potential LIP, herein named the "Hartley Event", is now recognized. Included in the potential Hartley LIP are previously dated 1.93–1.91 Ga magmatic units of the Kaapvaal Craton (i.e., Moshaneng sills and Trompsburg Complex). In addition, the new palaeomagnetic data presented in this paper from the Hartley Formation are combined with previously published results to define an improved 1.93–1.91 Ga palaeopole for the Hartley LIP that assists in refining the late Palaeoproterozoic APW path of the Kaapvaal Craton.

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References

Armstrong, R.A., 1987: *Geochronological Studies on Archaean and Proterozoic Formations of the Foreland of the Namaqua Front and Possible Correlates on the Kaapvaal Craton*. Unpublished Ph. D thesis. University of the Witwatersrand, Johannesburg, South Africa. 273 pp.

Barker, O.B., 1983: A proposed geotectonic model for the Soutpansberg Group within the Limpopo Mobile Belt, South Africa. *In* W.J. Van Biljon & J.H. Legg (eds.): *The Limpopo Belt. Special Publications of the Geological Society of South Africa, No. 8*, 181–190. The Geological Society of South Africa, Johannesburg, South Africa.

Barton, E.S. & Burger, A.J., 1983: Reconnaissance isotope investigations in the Namaqua Mobile Belt and implications for Proterozoic crustal evolution – Uppington geotraverse. *In* B.J.V. Botha (ed.): *Namaqualand Metamorphic Complex. Special Publication of the Geological Society of South Africa, No. 10*, 173–191. The Geological Society of South Africa, Johannesburg.

Barton, J.M., 1979: The chemical compositions, Rb-Sr isotopic systematics and tectonic setting of certain post-kinematic mafic igneous rocks, Limpopo Mobile Belt, Southern Africa. *Precambrian Research 9*, 57–80.

Beukes, N.J., Dorland, H.C., Gutzmer, J., Nedachi, M. & Ohmoto, H., 2002: Tropical laterites, life on land, and the history of atmospheric oxygen in the Paleoproterozoic. *Geology 30*, 491–494.

Bleeker, W. & Ernst, R.E., 2006: Short-lived mantle generated magmatic events and theri dike swarms: the key unlocking Earth's paleogeographic record back to 2.6 Ga *In:* E. Hanski, S. Mertanen, T. Rämö & J. Vuollo (eds.): *Dike Swarms: Time Markers of Crustal Evolution*, 3–26. Taylor & Francis/Balkema, Leiden.

Briden, J.C., Duff, B.A. & Kröner, A., 1979: Palaeomagnetism of the Koras group, Northern Cape province, South Africa. *Precambrian Research 10*, 43–57.

Carpozen, L., Gilder, S.A. & Hart, R.J., 2006: Origin and implications of two Verwey transitions in the basement rocks of the Vredefort meteorite crater, South Africa. *Earth and Planetary Science Letters 251*, 305–317.

Coffin, M.F. & Eldholm, O., 1993: Scratching the surface: estimating dimensions of large igneous provinces. *Geology 21*, 515–518.

Compston, W. & Williams, I.S., 1984: U–Pb geochronology of zircons form lunar Breccia 73217 using a sensitive high mass-resolution ion microprobe. *Journal of Geophysical Research 89*, B525–B534.

Compston, W., Williams, I.S., Kirschvink, J.L., Zang, Z. & Ma, G., 1992: Zircon U–Pb ages for the Early Cambrian timescale. *Journal of the Royal Geological Society, London 149*, 171–184.

Corcoran, P.L., Bumby, A.J. & Davis, D.W., 2013: The Paleoproterozoic Waterberg Group, South Africa: provenance and its relation to the timing of the Limpopo orogeny. *Precambrian Research 230*, 45–60.

Cornell, D.H., 1987: A field and petrographic study of the Hartley Basalt Formation, Northern Cape Province of South Africa. *Transactions Geological Society South Africa 90*, 5–27.

Cornell, D.H., Armstrong, R.A. & Walraven, F., 1998: Geochronology of the Proterozoic Hartley Basalt formation, South Africa: constraints on the Kheis tectogenesis and the Kaapvaal Craton's earliest Wilson Cycle. *Journal of African Earth Sciences 26*, 5–27.

Cornell, D.H., Thomas, R.J. & Moen, H.F.G., 2006: The Namaqua-Natal Province. *In* M.R. Johnson, C.R. Anhaeusser & R.J. Thomas (eds.): *The Geology of South Africa*, 325–379. Geological Society of South Africa/Council of Geoscience, Johannesburg/Pretoria.

Da Silva, R., 2011: *Distribution and Geochronology of Unconformity-bounded Sequences in Paleoproterozoic Elim-Olifansthoek Red Beds: Implications for Timing of Formation of Shishen-type Iron Ore and Heavy Carbonate Carbon Isotope Excursion*. Unpublished MSc thesis. University of Johannesburg. 103 pp.

de Kock, M.O., Beukes, N.J. & Armstrong, R.A., 2012: New SHRIMP U–Pb zircon ages from the Hartswater Group, South Africa: implications for correlations of the Neoarchean Ventersdorp Supergroup on the Kaapvaal craton and with the Fortesque Group on the Pilbara craton. *Precambrian Research 204–205*, 66–74.

de Kock, M.O., Ernst, R.E., Söderlund, U., Jourdan, F., Hofmann, A., Le Gall, B., Bertrand, H., Chisonga, B.C., Beukes, N.J., Rajesh, H.M., Moseki, L.M. & Fuchs, R., 2014: Dykes of the 1.11 Ga Umkondo LIP, Southern Africa: clues to a complex plumbing system. *Precambrian Research 249*, 129–143.

de Kock, M.O., Evans, D.A.D., Dorland, H.C., Beukes, N.J. & Gutzmer, J., 2006: Paleomagnetism of the lower two unconformity-bounded sequences of the Waterberg Group, South Africa: towards a better-defined apparent polar wander path for the Paleoproterozoic Kaapvaal Craton. *South African Journal of Geology 109*, 157–182.

Dorland, H.C., Beukes, N.J., Gutzmer, J., Evans, D.A.D. & Armstrong, R.A., 2006: Precise SHRIMP U–Pb zircon age constraints on the lower Waterberg and Soutpansberg Groups, South Africa. *South African Journal of Geology 109*, 139–156.

Eglington, B.M., 2006: Evolution of the Namaqua-Natal Belt, southern Africa – a geochronological and isotope geochemical review. *Journal of African Earth Sciences 46*, 93–111.

Eglington, B.M. & Armstrong, R.A., 2004: The Kaapvaal Craton and adjacent orogens, southern Africa: a geochronological database and overview of the geological development of the craton. *South African Journal of Geology 107*, 13–32.

Eriksson, P.G., Bumby, A.J., Brümer, J.J. & van der Neut, M., 2006: Precambrian fluvial deposits: enigmatic palaeohydrological data from the c. 2–1.9 Ga Waterberg Group, South Africa. *Sedimentary Geology 190*, 25–46.

Ermanovics, I., Key, R.M. & Jones, M.T., 1978: The Palapye Group, central-eastern Botswana. *Transactions of the Geological Society of South Africa 81*, 61–73.

Ernst, R.E., 2014: *Large Igneous Provinces*. Cambridge University Press, Cambridge.

Evans, D.A., Beukes, N.J. & Kirschvink, J.L. 1997: Low-latitude glaciation in the Palaeoproterozoic era. *Nature* 386, 262–266.

Evans, D.A.D., Beukes, N.J. & Kirschvink, J.L., 2002: Palaeomagnetism of a lateritic paleoweathering horizon and overlying Paleoproterozoic red beds from South Africa: implications for the Kaapvaal apparent polar wander path and a confirmation of atmospheric oxygen enrichment. *Journal of Geophysical Research-Solid Earth and Planets 107*, 1–22.

Fairey, B., Tsikos, H., Corfu, F. & Polteau, S., 2013: U–Pb systematics in carbonates of the Postmasburg Group, Transvaal Supergroup, South Africa: primary versus metasomatic controls. *Precambrian Research 231*, 194–205.

Geng, H., Brandl, G., Sun, M., Wong, J. & Kröner, A., 2014: Zircon ages defining deposition of the Palaeoproterozoic Soutpansberg Group and further evidence for Eoarchaean crust in South Africa. *Precambrian Research 249*, 247–262.

Goldberg, A.S., 2010: Dyke swarms as indicators of major extensional events in the 1.9–1.2 Ga Columbia supercontinent. *Journal of Geodynamics 50*, 176–190.

Gose, G.A., Hanson, R.E., Dalziel, I.W.D., Pancake, J.A. & Seidel, E.K., 2006: Paleomagnetism of the 1.1 Ga Umkondo large igneous province in southern africa. *Journal of Geophysical Research 111*, 1–18.

Gose, W.A., Hanson, R.E., Harmer, R.E. & Seidel, E.K., 2013: Reconnaissance paleomagnetic studies of Mesoproterozoic alkaline igneous complexes in the Kaapvaal craton, South Africa. *Journal of African Earth Sciences 85*, 22–30.

Gose, W.A., Johnston, S.T. & Thomas, R.J., 2004: Age of magnetization of Mesoproterozoic rocks from the Natal sector of the Namaqua-Natal belt, South Africa. *Journal of African Earth Sciences 40*, 137–145.

Gumsley, A.P., Bleeker, W., Chamberlain, K., Söderlund, U., de Kock, M.O., Kampmann, T. & Larsson, E., 2015: U–Pb TIMS and *in situ* SIMS dating of baddeleyite and zircon from sub-volcanic sills of the Ongeluk Formation (Transvaal SupergroupP in the Griqualand West sub-basin, Kaapvaal Craton, with implications for Snowball Earth and the Great Oxygenation Event. *Joint Assembly of the AGU, GAC-MAC-CGU*, 378–379. 3–7 May 2015, Montreal, Canada.

Gumsley, A.P., de Kock, M.O., Rajesh, H.M., Knoper, M.W., Söderlund, U. & Ernst, R.E., 2013: The Hlagothi Complex: the identification of fragments from a Mesoarchean large igneous province on the Kaapvaal Craton. *Lithos 174*, 333–348.

Gumsley, A.P., Olsson, J.R., Söderlund, U., de Kock, M.O., Hofmann, A. & Klausen, M., 2015: Precise U–Pb baddeleyite age dating of the Ushushwana Complex, southern Africa – implications for the Mesoarchaean magmatic and sedimentological evolution of the Pongola Supergroup, Kaapvaal Craton. *Precambrian Research 257*, 174–185.

Hanson, R.E., Gose, W.A., Crowley, J.L., Ramezani, J., Bowring, S.A., Bullen, D.S., Hall, R.P., Pancake, J.A. & Mukwakwami, J., 2004: Paleoproterozoic intraplate magmatism and basin development on the Kaapvaal Craton: age,

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paleomagnetism and geochemistry of ~1.93 to ~1.87 Ga post-Waterberg dolerites. *South African Journal of Geology 107*, 233–254.

Hanson, R.E., Harmer, R.E., Blenkinsop, T.G., Bullen, D.S., Dalziel, I.W.D., Gose, W.A., Hall, R.P., Kampunzu, A.B., Key, R.M., Mukwakwami, J., Munyanyiwa, H., Pancake, J.A., Seidel, E.K. & Ward, S.E., 2006: Mesoproterozoic intraplate magmatism in the Kalahari Craton: a review. *Journal of African Earth Sciences 46*, 141–167.

Hanson, R.E., Rioux, M., Gose, W.A., Blackburn, T.J., Bowring, S.A., Mukwakwami, J. & Jones, D.L., 2011: Paleomagnetic and geochronological evidence for large-scale post 1.88 Ga displacement between the Zimbabwe and Kaapvaal cratons along the Limpopo belt. *Geology 39*, 487–490.

Hargraves, R.B. & Onstott, T.C., 1980: Paleomagnetic results from some southern African kimberlites, and their tectonic significance. *Journal of Geophysical Research 85*, 3587–3596.

Jaffey, A.H., Flynn, K.F., Glendenin, L.E., Bently, W.C. & Essling, A.M., 1971: Precision measurement of half-lives and specific activities of U235 and U238. *Physical Review C 4*, 1889–1906.

Jones, C.H., 2002: User-driven integrated software lives: "Paleomag" paleomagnetic analysis on the Macintosh. *Computers & Geosciences 28*, 1145–1115.

Kamo, S.L., Reimold, W.U., Krogh, T.E. & Colliston, W.P., 1996: A 2.023 Ga age for the Vredefort impact event and a first report of shock metmorphosed zircons in pseudotachylitic breccias and granophyre. *Earth and Planetary Science Letters 144*, 369–387.

Kampmann, T., Gumsley, A.P., de Kock, M.O. & Söderlund, U., 2015: U–Pb geochronology and paleomagnetism of the Westerberg Sill Suite, Kaapvaal Craton – support for a coherent Kaapvaal–Pilbara Block (Vaalbara) into the Paleoproterozoic? *Precambrian Research 269*, 58–72.

Kirschvink, J.L., 1980: The least-squares line and plane and the analysis of palaeomagnetic data. *Geophysical Jounral of the Royal Astronomical Society 62*, 699–718.

Kirschvink, J.L., Kopp, R.E., Raub, T.D., Baumgartner, C.T. & Holt, J.W., 2008: Rapid, precise, and high-sensitivity acquisition of paleomagnetic and rock-magnetic data: development of a low-noise automatic sample changing system for superconducting rock magnetometers. *Geochemistry Geophysics Geosystems 9*, 1–18.

Kramers, J.D. & Mouri, H., 2011: The geochronology of the Limpopo Complex: a controversy solved. *In* D.D. Van Reenen, J.D. Kramers, S. McCourt & L.L. Perchuk (eds.): *Origin and Evolution of Precambrian High-Grade Gneiss Terranes, with Special Emphasis on the Limpopo Complex of South Africa: Geological Society of America Memoir 207*, 85–106. The Geological Society of America.

Layer, P.W., Kröner, A., McWilliams, M. & Clauer, N., 1988: Regional magnetic overprinting of Witwatersrand Supergroup Sediments, South Africa. *Journal of Geophysical Research 93*, 2191–2200.

Letts, S., Torsvik, T.H., Webb, S.J. & Ashwal, L.D., 2009: Palaeomagnetism of the 2054 Ma Bushveld Complex (South Africa): implications for emplacement and cooling. *Geophysical Journal International 179*, 850–872.

Letts, S., Torsvik, T.H., Webb, S.J. & Ashwal, L.D., 2010: New Palaeoproterozoic palaeomagnetic data from the Kaapvaal Craton, South Africa. *Geological Society, London, Special Publications 357*, 9–26.

Lubnina, N., Ernst, R.E., Klausen, M. & Söderlund, U., 2010: Paleomagnetic study of NeoArchean–Paleoproterozoic dykes in the Kaapvaal Craton. *Precambrian Research 183*, 523–552.

Ludwig, K.R., 1991: Isoplot – a plotting and regression program for radiogenic isotopic data. *Open File Report*. USGS. 445 pp.

Maier, W.D., Peltonen, P., Grantham, G. & Mänttäri, I., 2003: A new 1.9 Ga age for the Trompsburg intrusion, South Africa. *Earth and Planetary Science Letters 212*, 351–360.

Mapeo, R.B.M., Ramokate, L.V., Armstrong, R.A. & Kampunzu, A.B., 2004: U–Pb zircon age of the upper Palapye group (Botswana) and regional implications. *Journal of African Earth Sciences 40*, 1–16.

Moen, H.F.G., 1999: The Kheis Tectonic Subprovince, southern Africa: a lithostratigraphic perspective. *South African Journal of Geology 102*, 27–42.

Moen, H.F.G., 2006: The Olifantshoek Supergroup. *In* M.R. Johnston, C.R. Anhaeusser & R.J. Thomas (eds.): *The Geology of South Africa*, 319–324. The Geological Society of South Africa and Council for Geoscience, Johannesburg and Pretoria.

Morgan, G.E., 1985: The paleomagnetism and cooling history of metamorphic and igneous rocks from the Limpopo Mobile Belt, southern Africa. *Geological Society of America Bulletin 96*, 663–675.

Morgan, G.E. & Briden, J.C., 1981: Aspects of Precambrian palaeomagnetism, with new data from the Limpopo Mobile Belt and Kaapvaal Craton in soouthern Africa. *Physics of the Earth and Planetary Interiors 24*, 142–168.

Mukasa, S.B., Wilson, A.H. & Young, K.R., 2013: Geochronological constraints on the magmatic and tectonic development of the Pongola Supergroup (Central Region), South Africa. *Precambrian Research 224*, 268–286.

Olsson, J.R., Klausen, M., Hamilton, M., März, N., Söderlund, U. & Roberts, R.J. 2015: Baddeleyite U-Pb ages and geochemistry of the 1875-1835 Ma Black Hills Dyke Swarm across north-eastern South Africa: part of a trans-Kalahari Craton back-arc setting? *GFF*. doi: http://dx.doi.org/10.1080/11035897.2015.1103781.

Olsson, J.R., Söderlund, U., Hamilton, M.A., Klausen, M.B. & Helffrich, G.R., 2011: A late Archean radiating dyke swarm as possible clue to the origin of the Bushveld Complex. *Nature Geoscience 4*, 865–869.

Olsson, J.R., Söderlund, U., Klausen, M.B. & Ernst, R.E., 2010: U–Pb baddeleyite ages linking major Archean dyke swarms to volcanic-rift forming events in the Kaapvaal craton (South Africa), and a precise age for the Bushveld Complex. *Precambrian Research 183*, 490–500.

Paces, J.B. & Miller, Jr, J.D., 1993: Precise U–Pb ages of Duluth Complex and related mafic intrusions, northeastern Minnesota: geochronological insights to physical, petrogenetic, paleomagnetic, and tectonomagmatic processes associated with the 1.1 Ga Midcontinent Rift System. *Journal of Geophysical Research 98*, 13997–14013.

Pettersson, Å., Cornell, D.H., Moen, H.F.G., Reddy, S. & Evans, D., 2007: Ionprobe dating of 1.2 Ga collision and crustal architecture in the Namaqua-Natal Province of southern Africa. *Precambrian Research 158*, 79–92.

Reischmann, T., 1995: Precise U/Pb age determination with baddeleyite (ZrO_2), a case study from the Palaborwa Igneous Complex, South Africa. *South African Journal of Geology 98*, 1–4.

Rotenberg, E., Davis, D.W., Amelin, Y., Gosh, S. & Bergquist, B.A., 2012: Determination of the decay-constant of ^{87}Rb by laboratory accumulation of ^{87}Sr. *Geochimica et Cosmochimica Acta 85*, 41–57.

Schütte, S.S., 1992: *Ongeluk Volcanism in Relation to the Kalahari Manganese Deposits*. Unpublished Ph. D. University of Natal. 262 pp.

Scoates, J.S. & Friedman, R.M., 2008: Presice age of the platineferous Merensky Reef, Bushveld Complex, South Africa, by the U–Pb zircon chemical abrasion ID-TIMS technique. *Economic Geology 103*, 465–471.

Smit, C.A., Beukes, N.J., Johnson, M.R., Malherbe, S.J. & Visser, J.N.J., 1991: *Lithostratigraphy of the Vryburg Formtion (Including the Kalkput, Geelbeksdam, Rosendal, Waterloo and Oceola Members). South African Committee for Stratigraphy Lithostratigraphic Series, No. 14*, Council for Geoscience, Pretoria.

Söderlund, U., Hofmann, A., Klausen, M.B., Olsson, J.R., Ernst, R.E. & Persson, P.-O., 2010: Towards a complete magmatic barcode for the Zimbabwe craton: Baddeleyite U–Pb dating of regional dolerite dyke swarms and sill complexes. *Precambrian Research 183*, 388–398.

Stacey, J.S. & Kramers, J.D., 1975: Approximation of terrestrial lead isotope evolution by a two-stage model. *Earth and Planetary Science Letters 26*, 207–221.

Tinker, J., De Wit, M.J. & Grotzinger, J., 2002: Seismic stratigraphic constraints on Neoarchean–Paleoproterozoic evolution of the western margin of the Kaapvaal Craton, South Africa. *South African Journal of Geology 105*, 107–134.

Tinker, J.H., De Wit, M.J. & Royden, L.H., 2004: Old, strong continental lithosphere with weak Archean margin at ~1.8 Ga, Kaapvaal Craton, South Africa. *South African Journal of Geology 107*, 255–260.

Van der Voo, R., 1990: The reliability of paleomagnetic data. *Tectonophysics 184*, 1–9.

Van Niekerk, H.S., 2006: *The Origin of the Kheis Terrane and its Relationship with the Archean Kaapvaal Craton and the Grenvillian Namaqua Province in Southern Africa*. Unpublished Ph. D thesis. University of Johannesburg. 241 pp.

Wabo, H., de Kock, M.O., Klausen, M., Söderlund, U. & Beukes, N.J. 2015a: *Paleomagnetism and chronology of B-1 marginal sills of the Bushveld Complex from the eastern Kaapvaal Craton*. South Africa, *GFF*. doi: http://dx.doi.org/10.1080/11035897.2015.1099566.

Wabo, H., Olsson, J.R., De Kock, M.O., Humbert, F., Söderlund, U. & Klausen, M. 2015b: New U-Pb age and paleomagnetic constraints from the UItkomst Complex. *South Africa: clues to the timing of intrusion. GFF.*. doi: http://dx.doi.org/10.1080/11035897.2015.1098726.

Williams, I.S. & Claesson, S., 1987: Isotopic evidence for the Precambrian provenance Caledonian metamorphism of high grade paragneisses from the Seve Nappes, Scandinavian Caledonides. II ion microbrobe zircon U–Th–Pb. *Contributions to Mineralogy and Petrology 97*, 205–217.

Williams, S.E., Müller, R.D., Landgrebe, T.C.W. & Whittaker, J.M., 2012: An open-source software environment for visualizing and refining plate tectonic reconstructions using high-resolution geological and geophysical data sets. *GSA Today 22*, 4–9.